



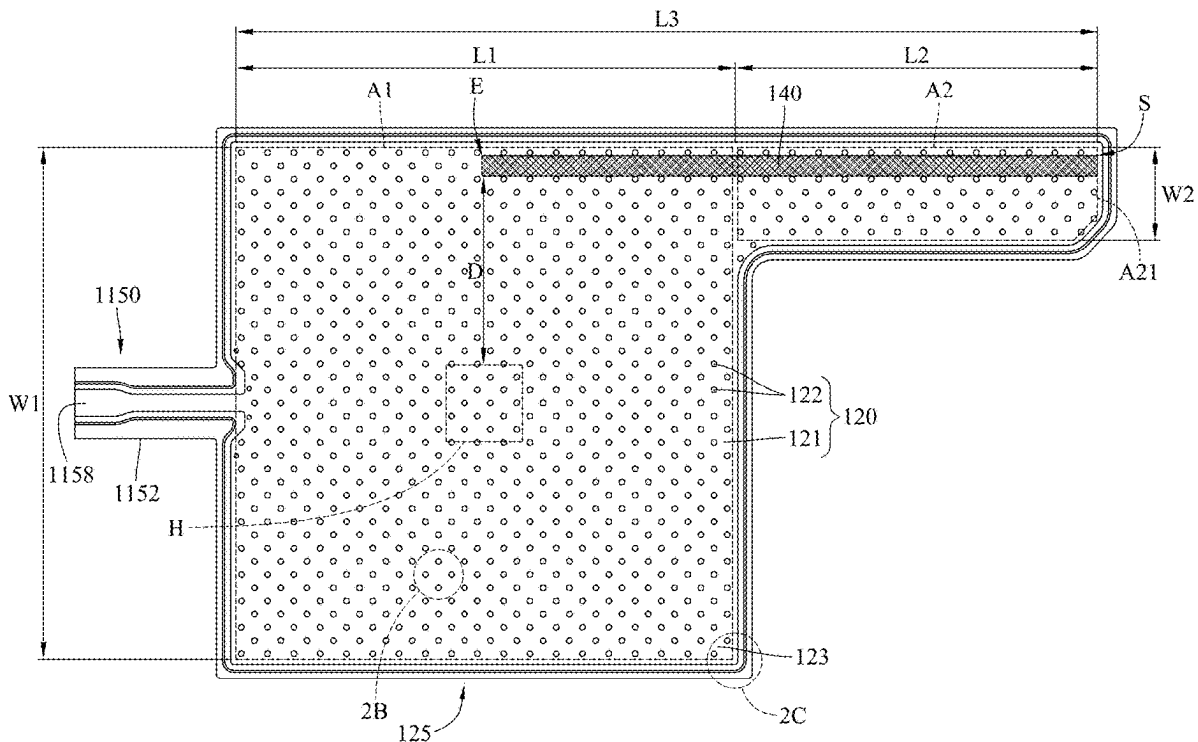
US 20250277630A1

(19) **United States**(12) **Patent Application Publication**
Wang et al.(10) **Pub. No.: US 2025/0277630 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **HEAT DISSIPATION DEVICE HAVING
IRREGULAR SHAPE**(30) **Foreign Application Priority Data**

Feb. 8, 2021 (CN) 202120353801.1

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Taipei City (TW)**Publication Classification**(72) Inventors: **Xuemei Wang**, Taipei City (TW);
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Jen-Chih Cheng, Taipei City (TW)(51) **Int. Cl.**
F28D 15/02 (2006.01)
F28D 15/04 (2006.01)
H05K 7/20 (2006.01)(73) Assignee: **COOLER MASTER CO., LTD.**,
Taipei City (TW)(52) **U.S. Cl.**
CPC **F28D 15/0233** (2013.01); **F28D 15/046**
(2013.01); **H05K 7/20336** (2013.01)(21) Appl. No.: **19/212,028**(22) Filed: **May 19, 2025**(57) **ABSTRACT****Related U.S. Application Data**(60) Division of application No. 17/665,440, filed on Feb.
4, 2022, now Pat. No. 12,331,997, which is a contin-
uation-in-part of application No. 16/711,233, filed on
Dec. 11, 2019, now Pat. No. 11,913,725.(60) Provisional application No. 62/798,480, filed on Jan.
30, 2019, provisional application No. 62/783,717,
filed on Dec. 21, 2018.

A heat dissipation device includes a first casing and a second casing coupled to the first casing. The second casing includes a body having an inner surface and an outer surface opposite the inner surface, and a first portion and a second portion, each of the first and second portions having a different cross-sectional area. The heat dissipation device further includes a plurality of columns on the inner surface, and a first wick structure disposed on the inner surface and in the first portion and the second portion.



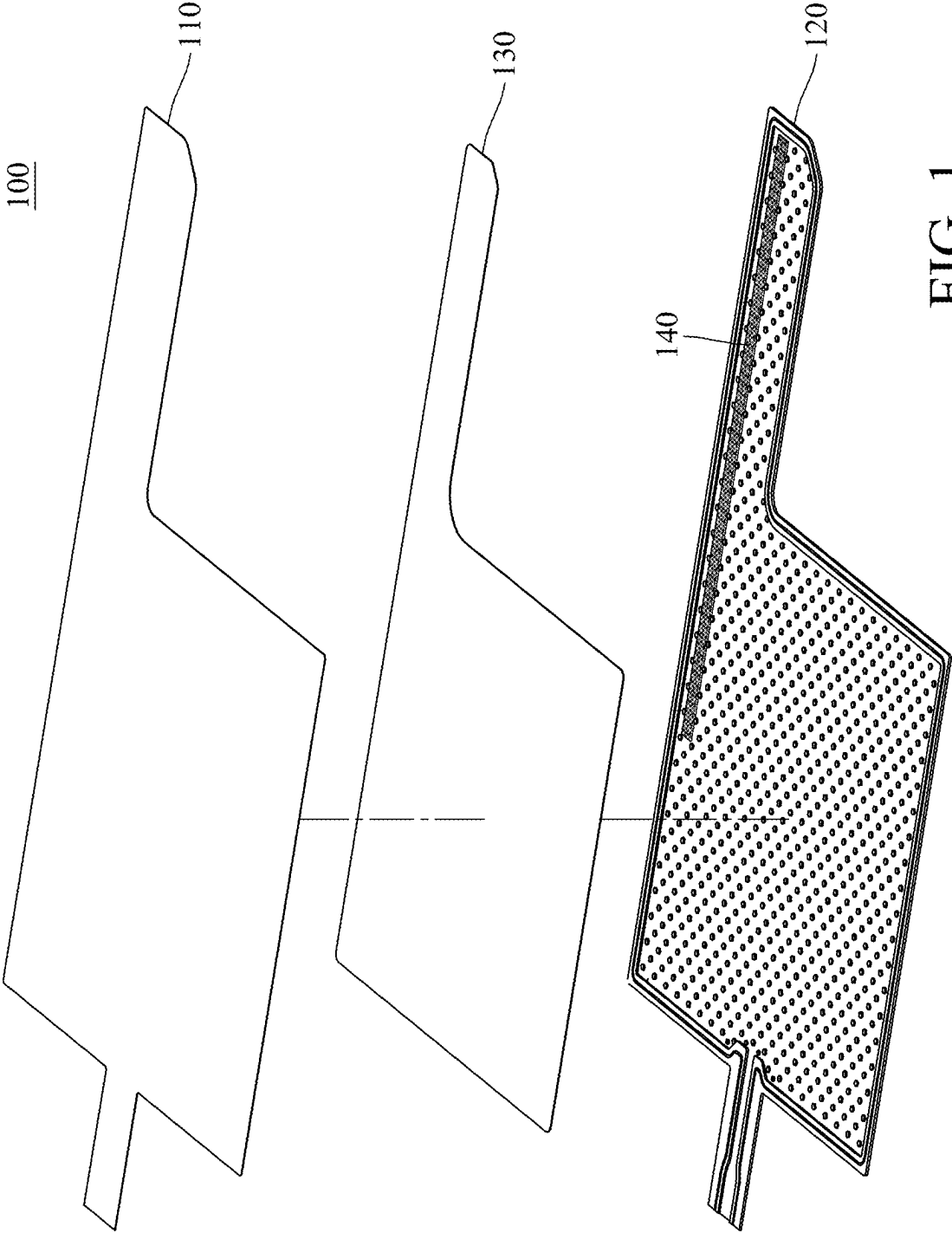
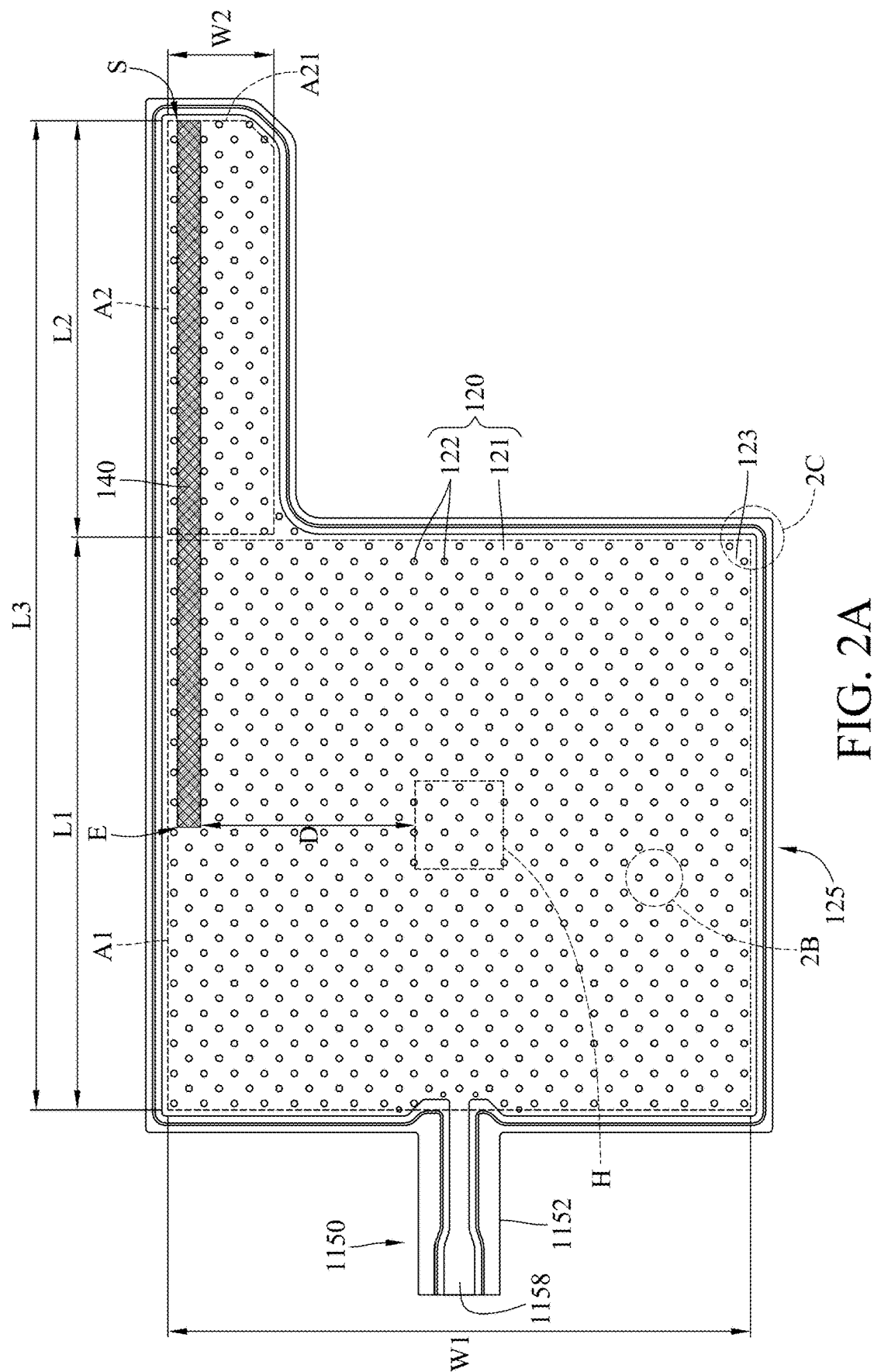


FIG. 1



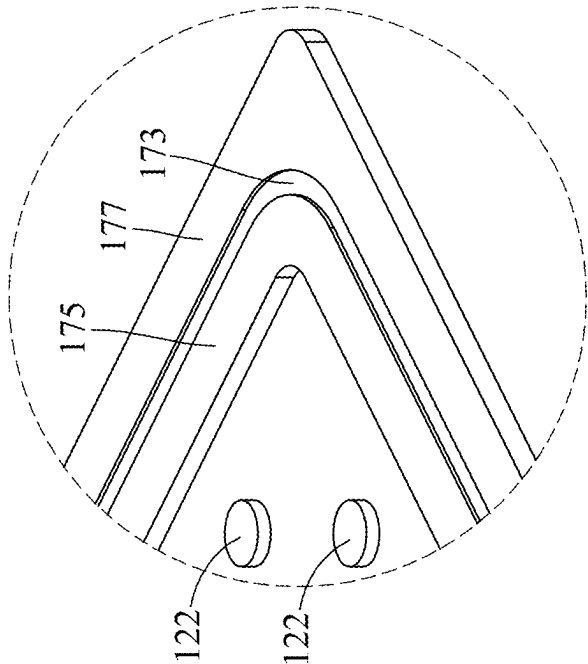


FIG. 2C

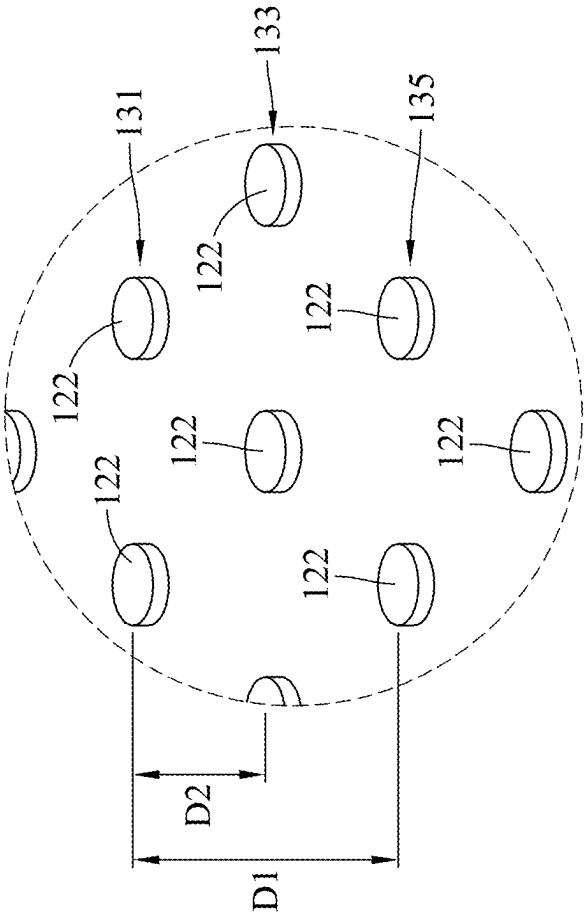


FIG. 2B

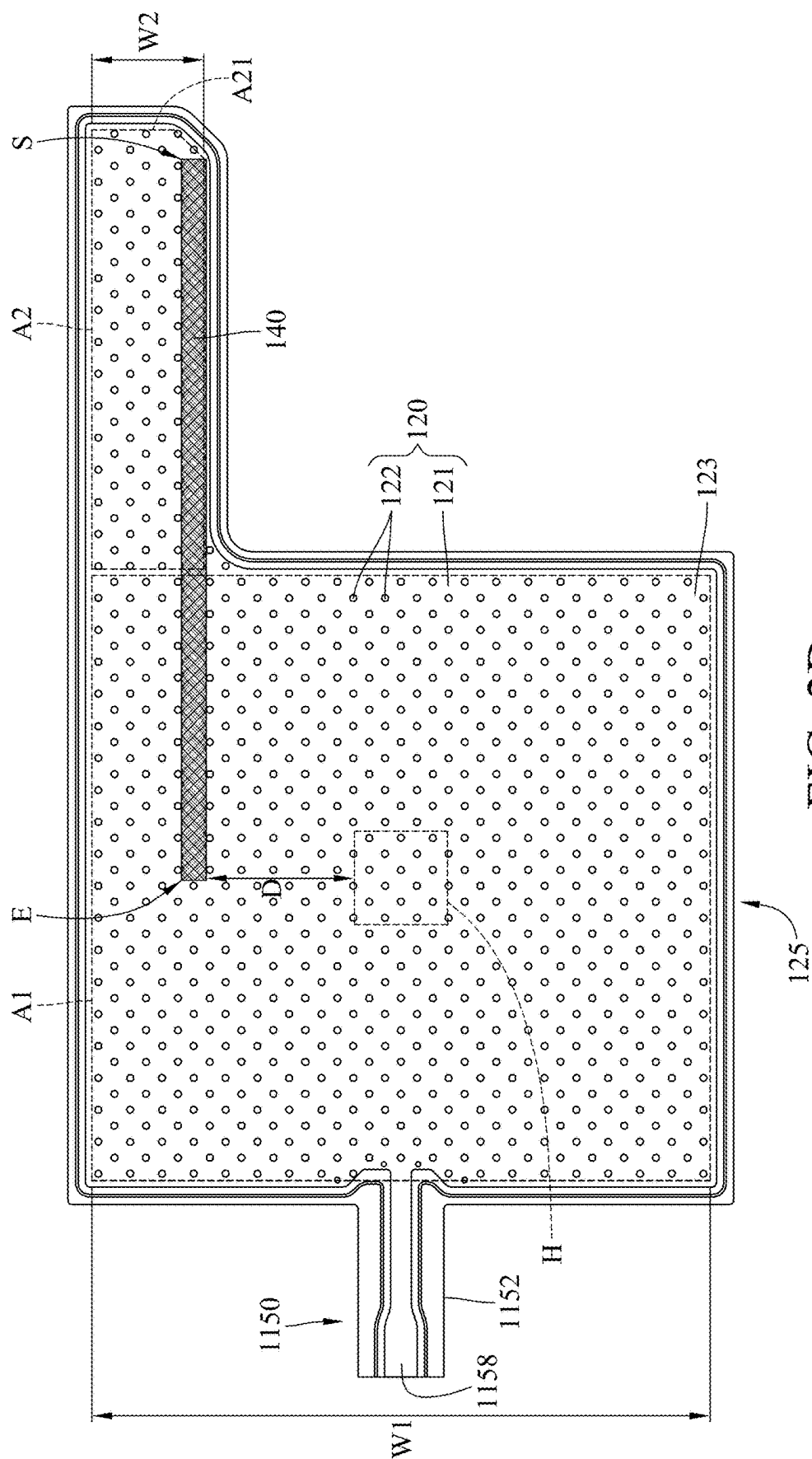


FIG. 2D

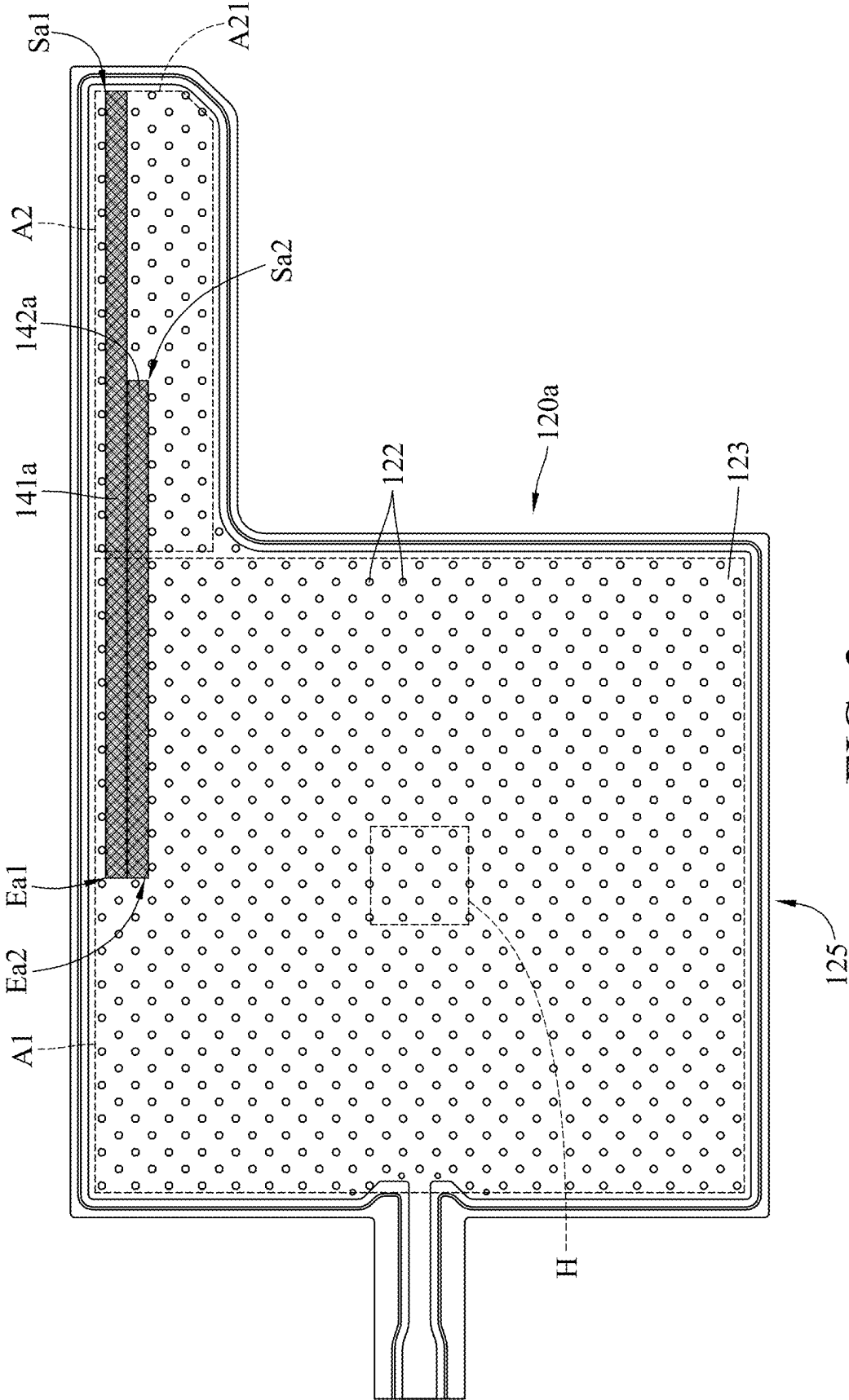


FIG. 3

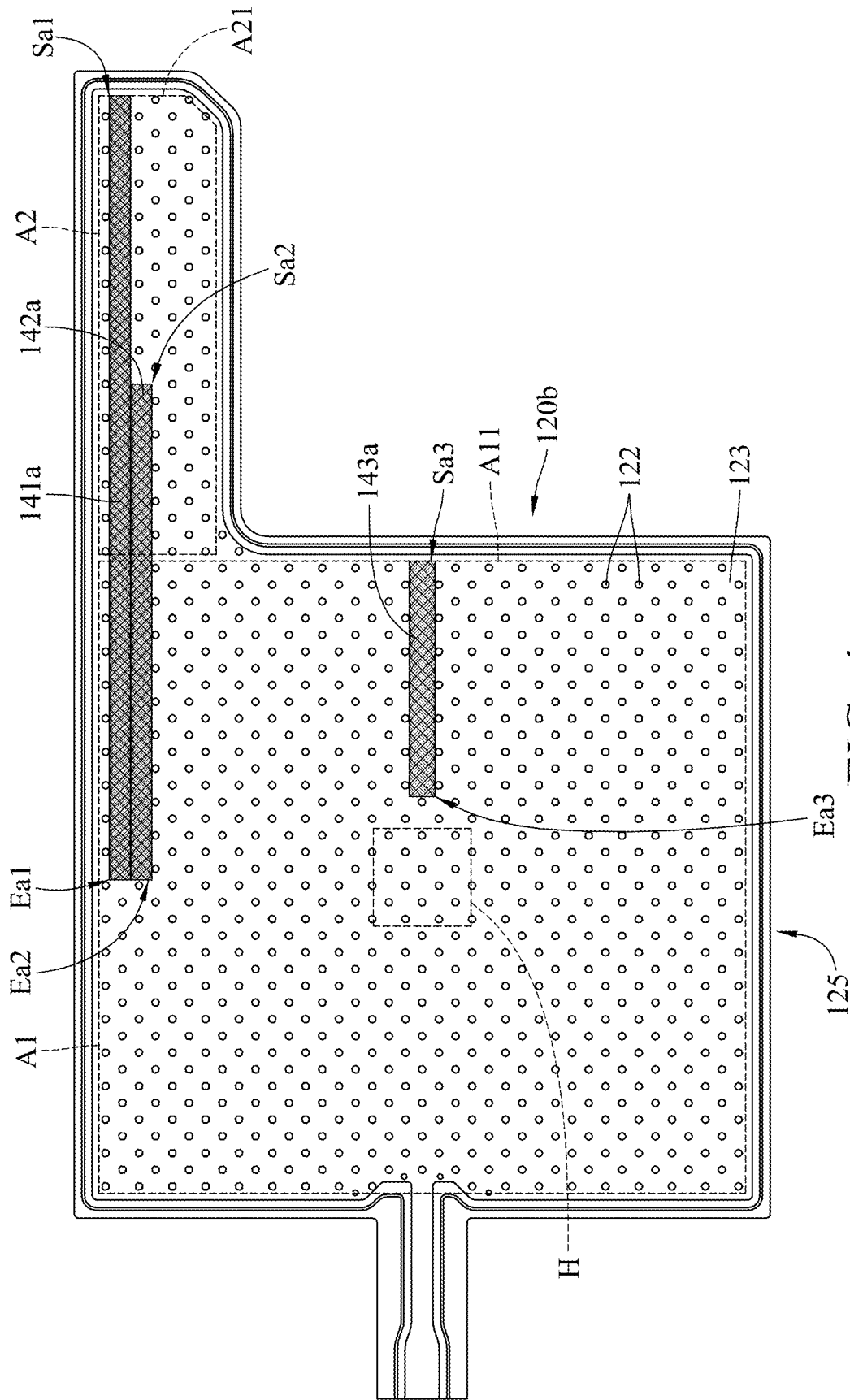
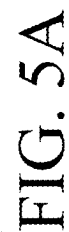


FIG. 4



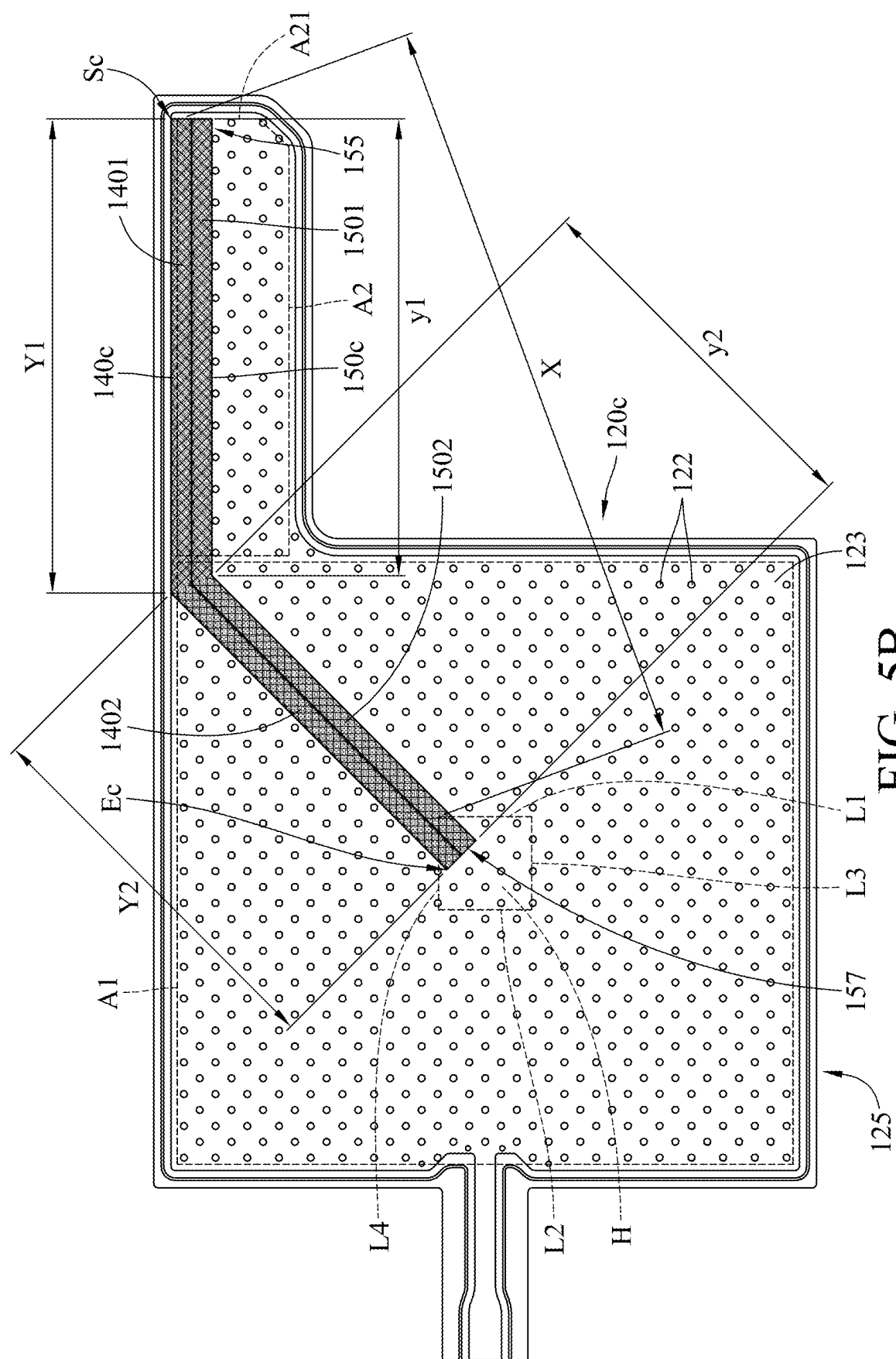
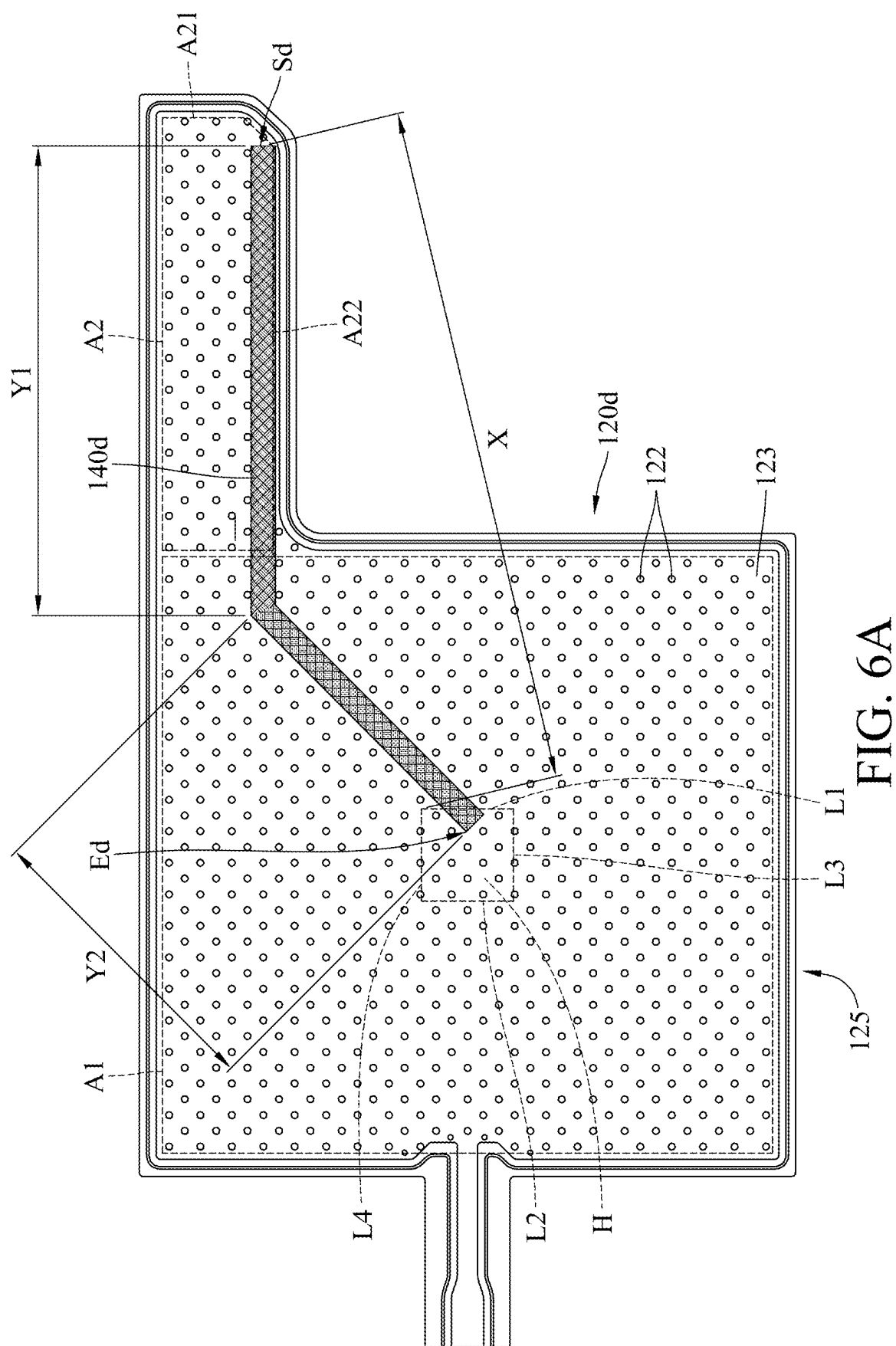
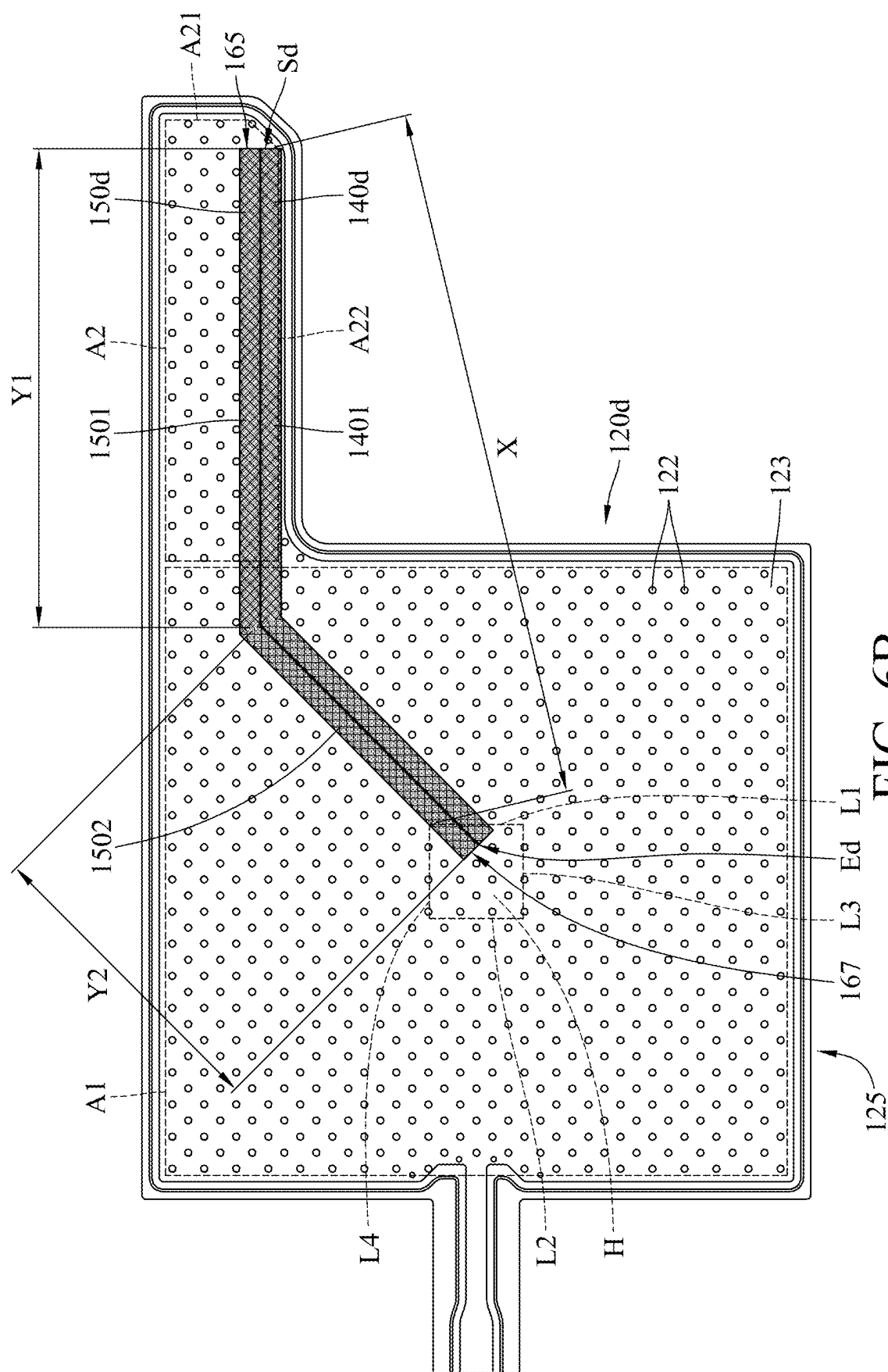


FIG. 5B





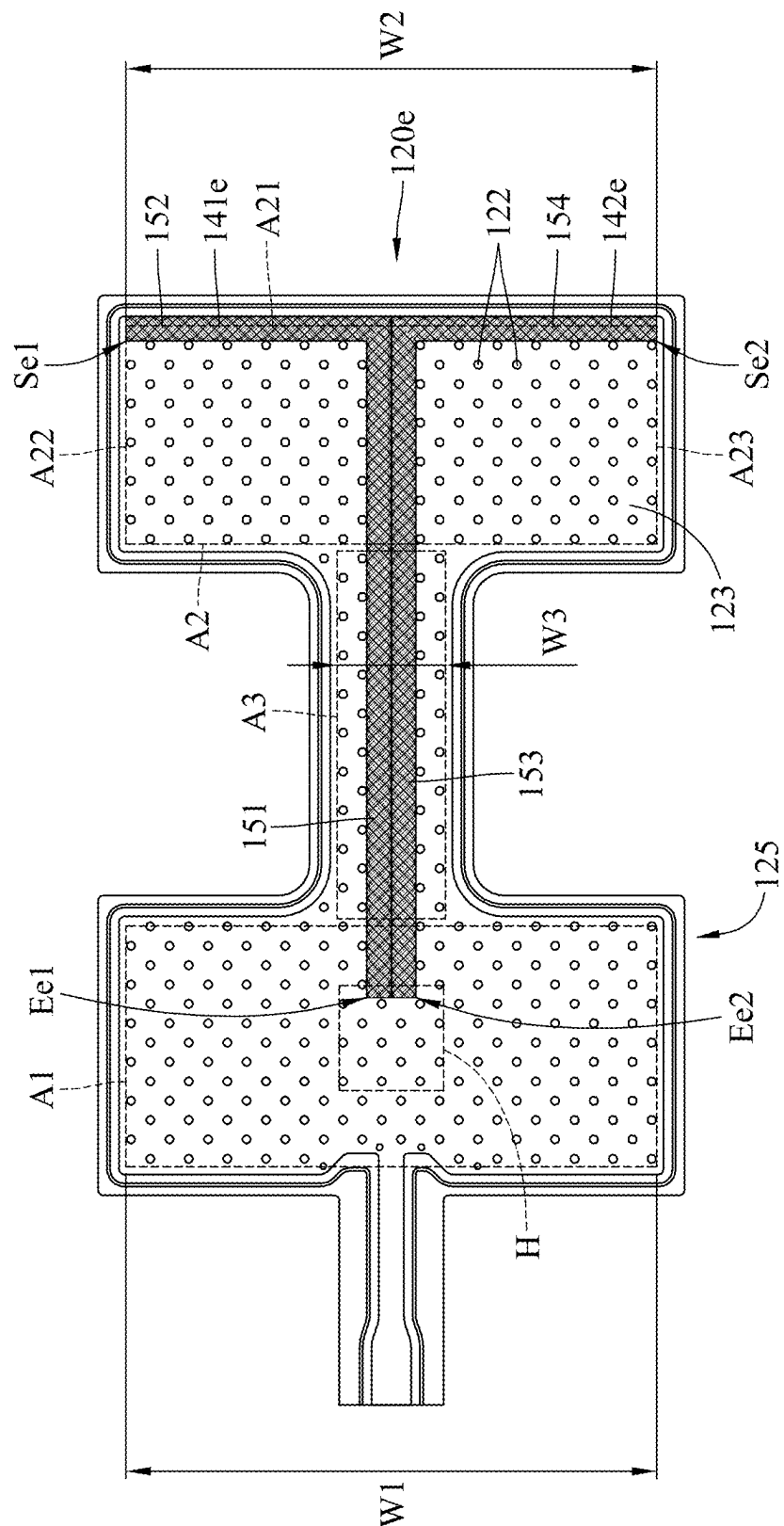


FIG. 7A

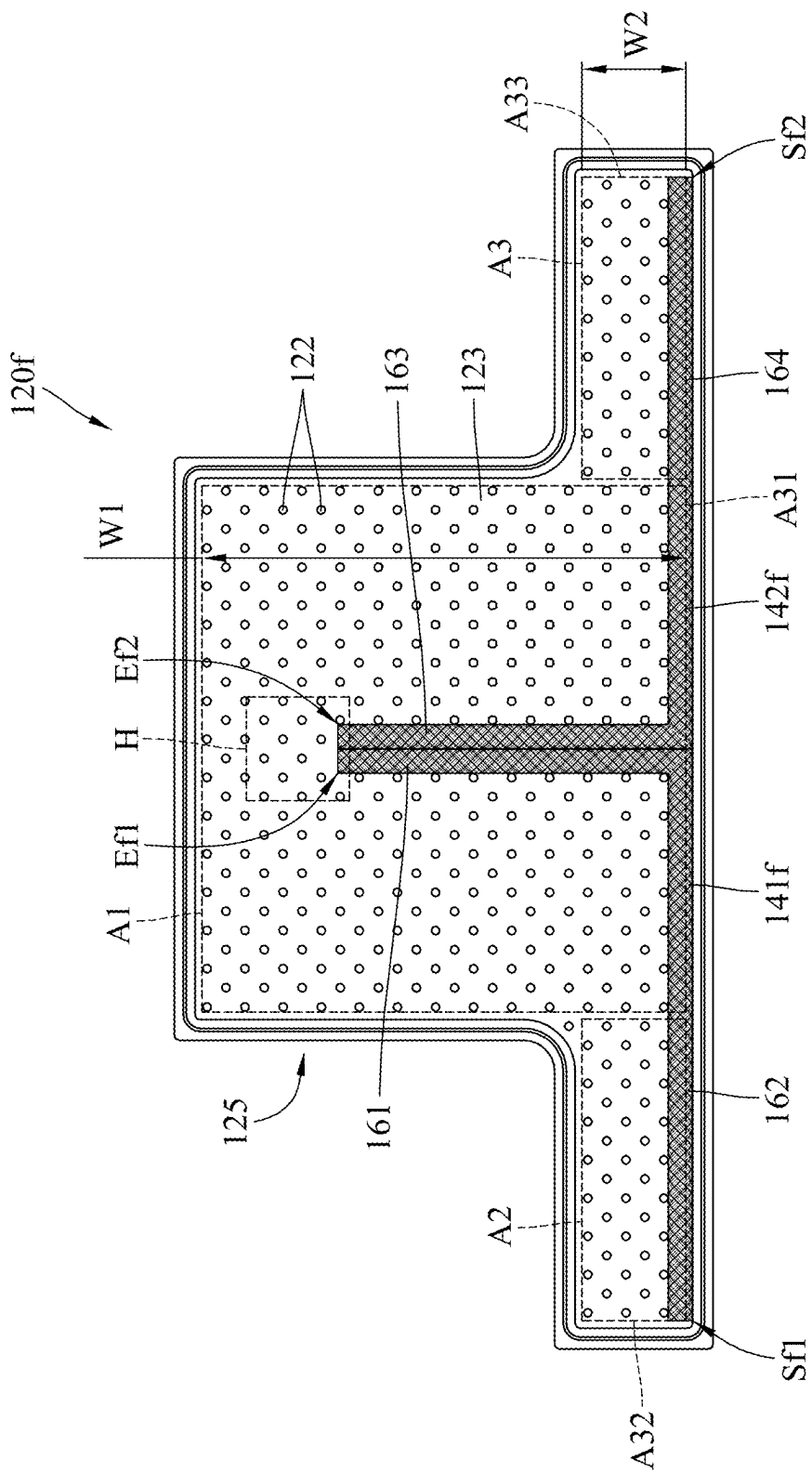


FIG. 8A

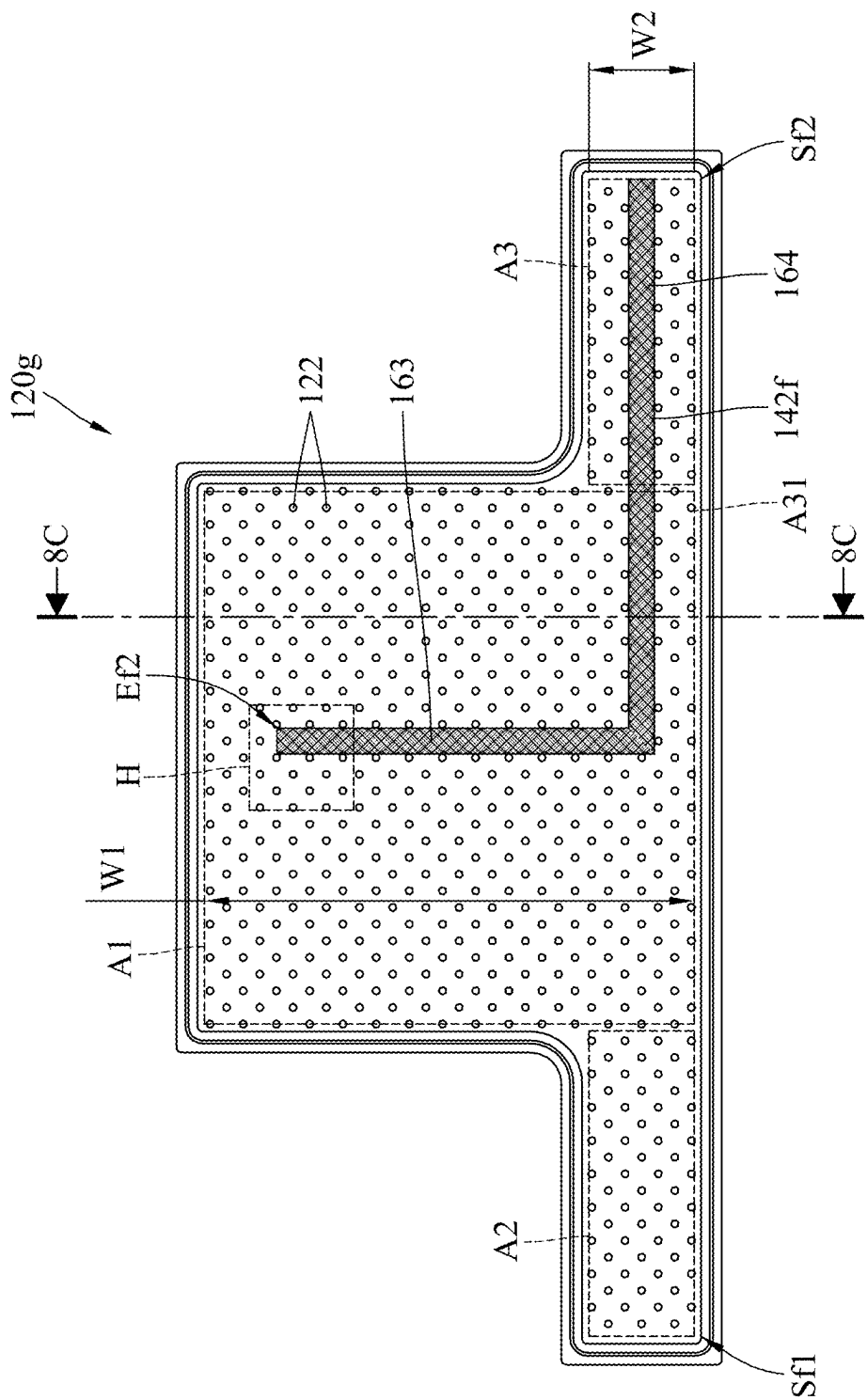


FIG. 8B

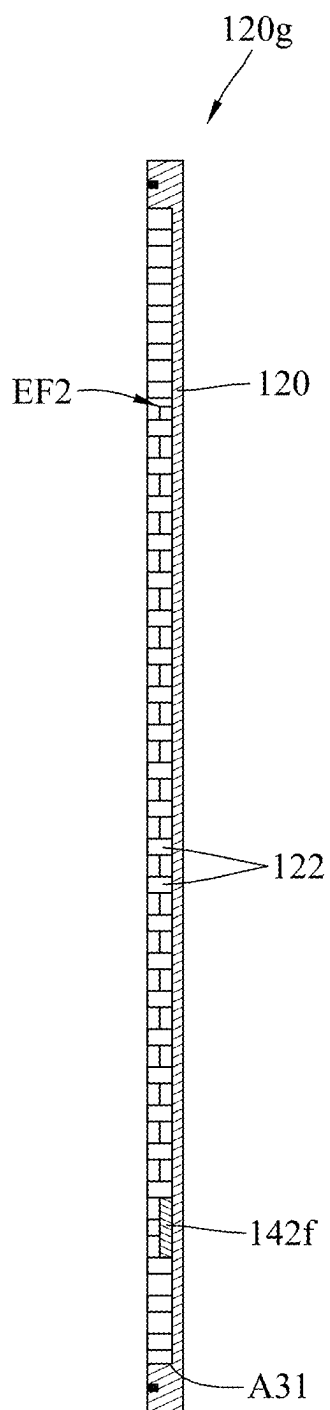
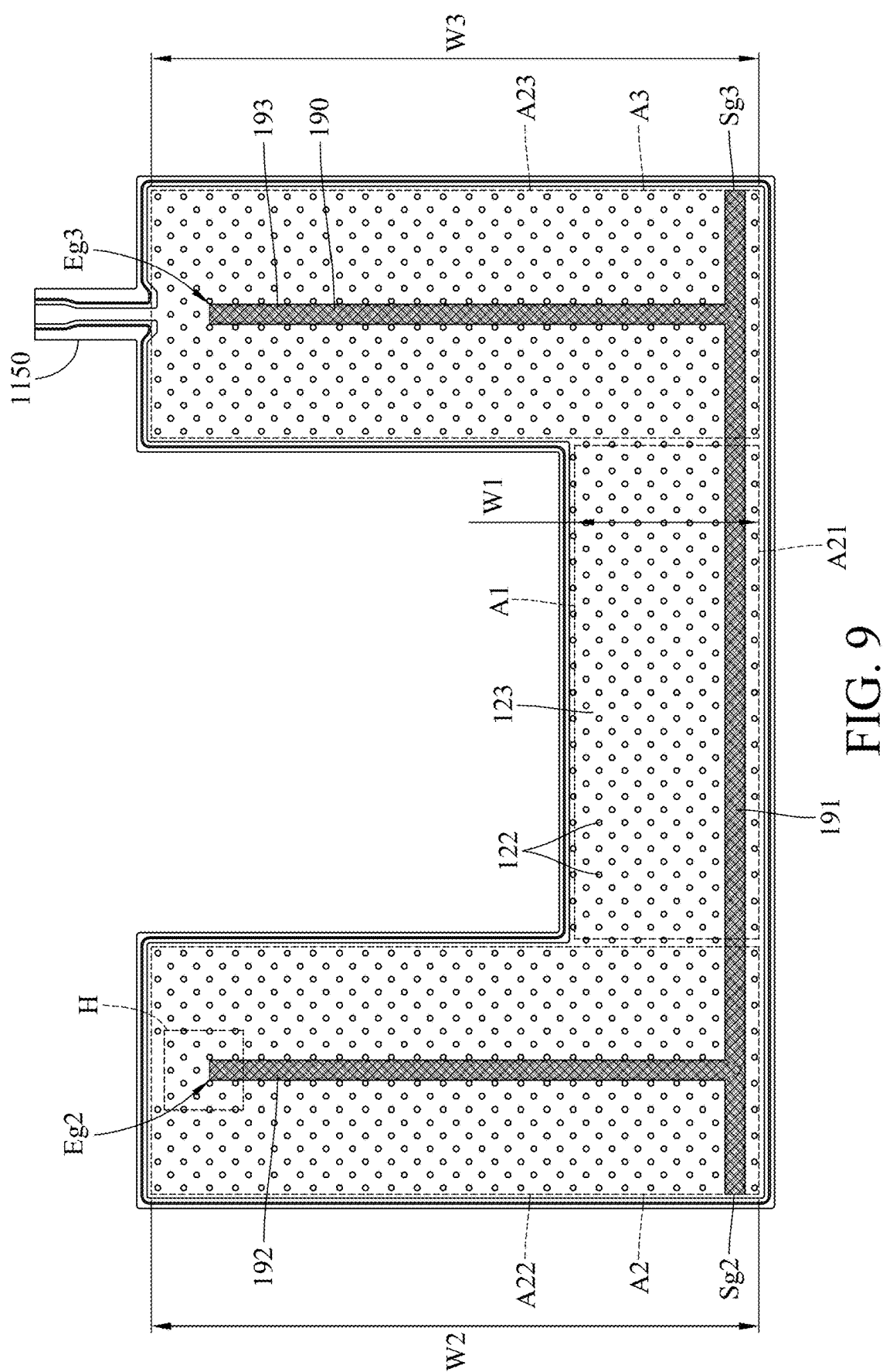


FIG. 8C



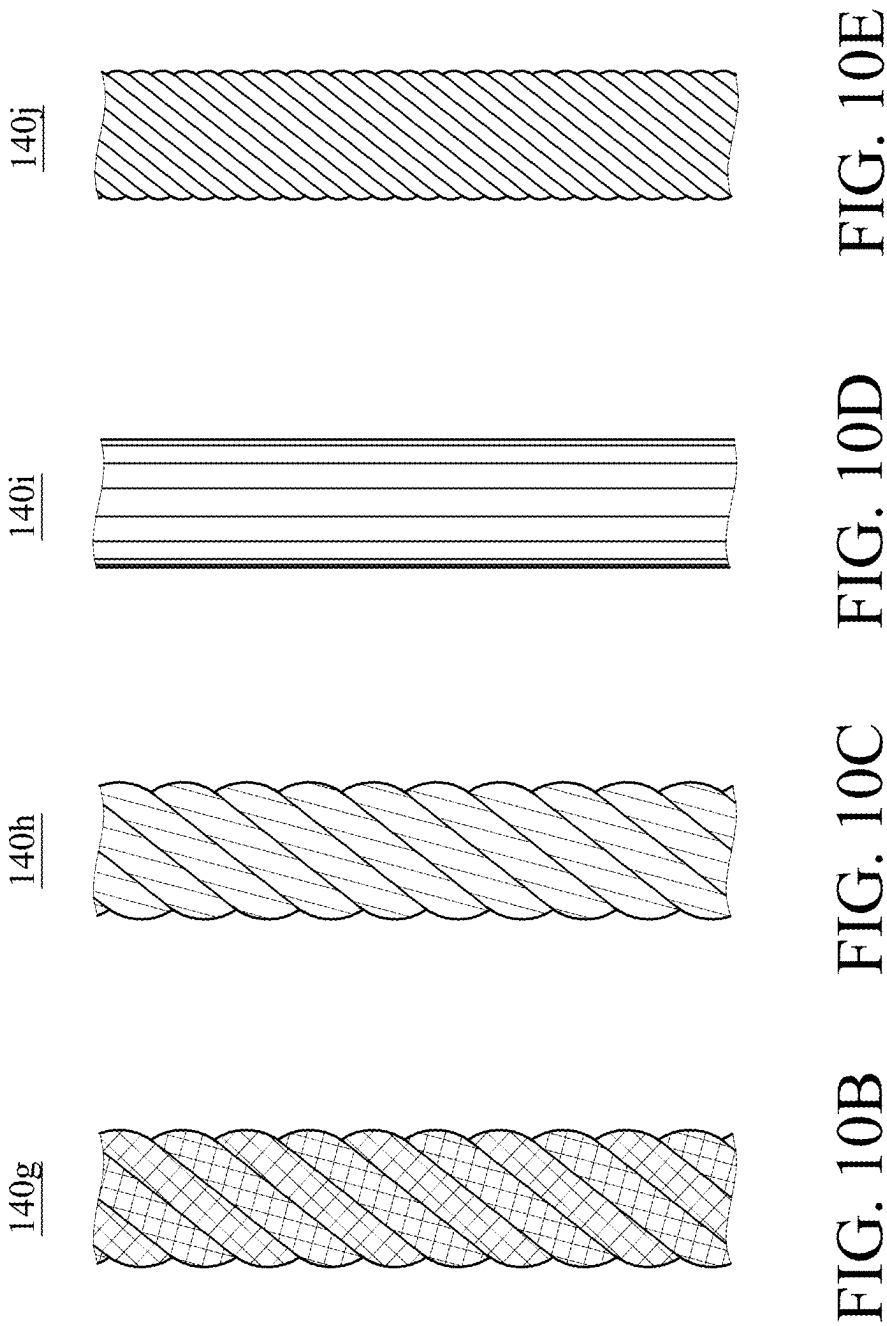


FIG. 10A

FIG. 10B

FIG. 10C

FIG. 10D

FIG. 10E

140k

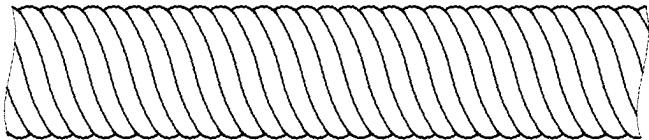


FIG. 10F

1030

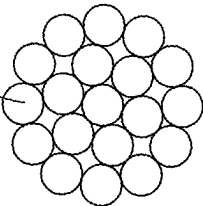


FIG. 10G

140m

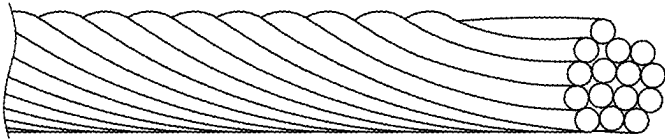


FIG. 10H

140n

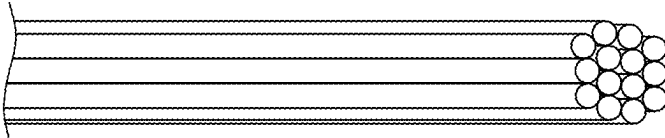


FIG. 10J

140p

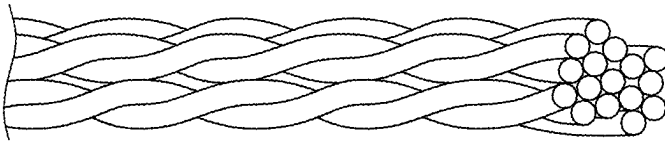


FIG. 10K

100i

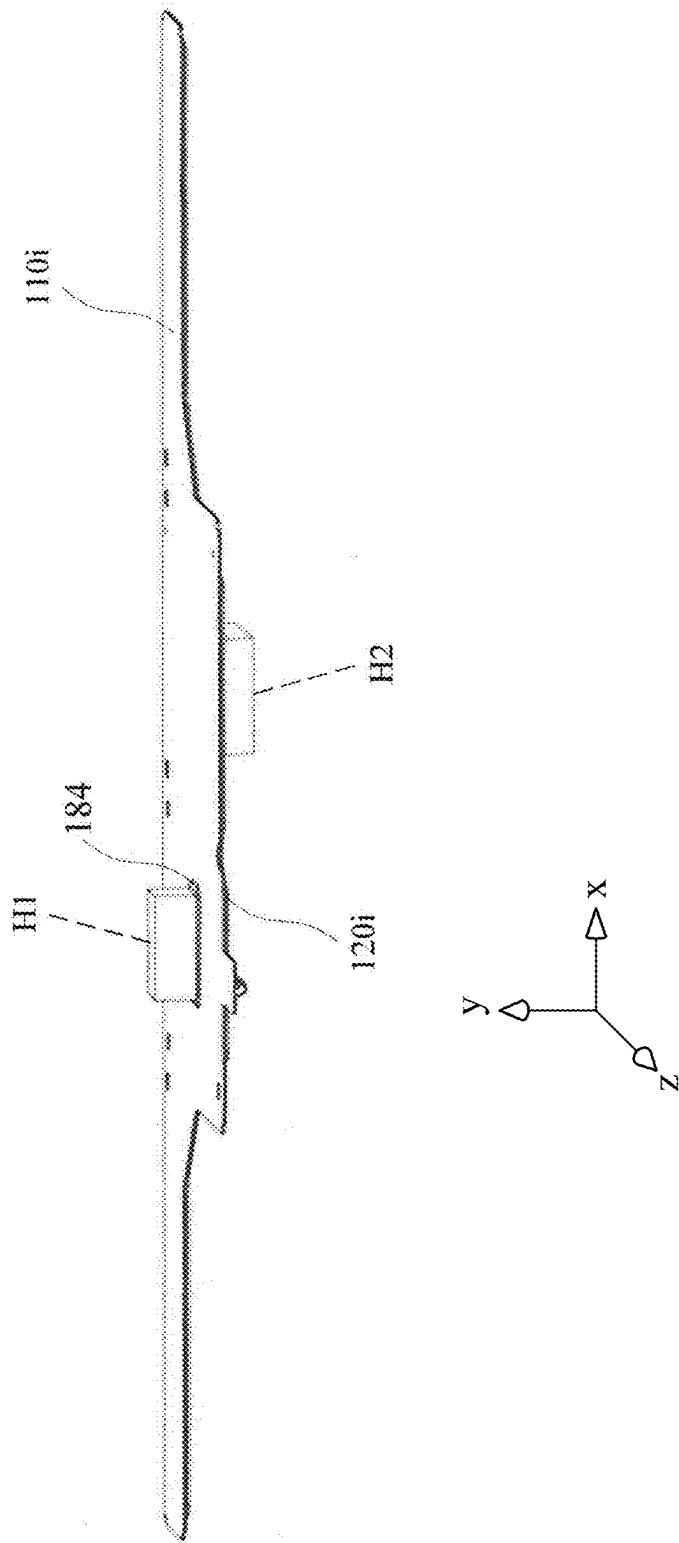
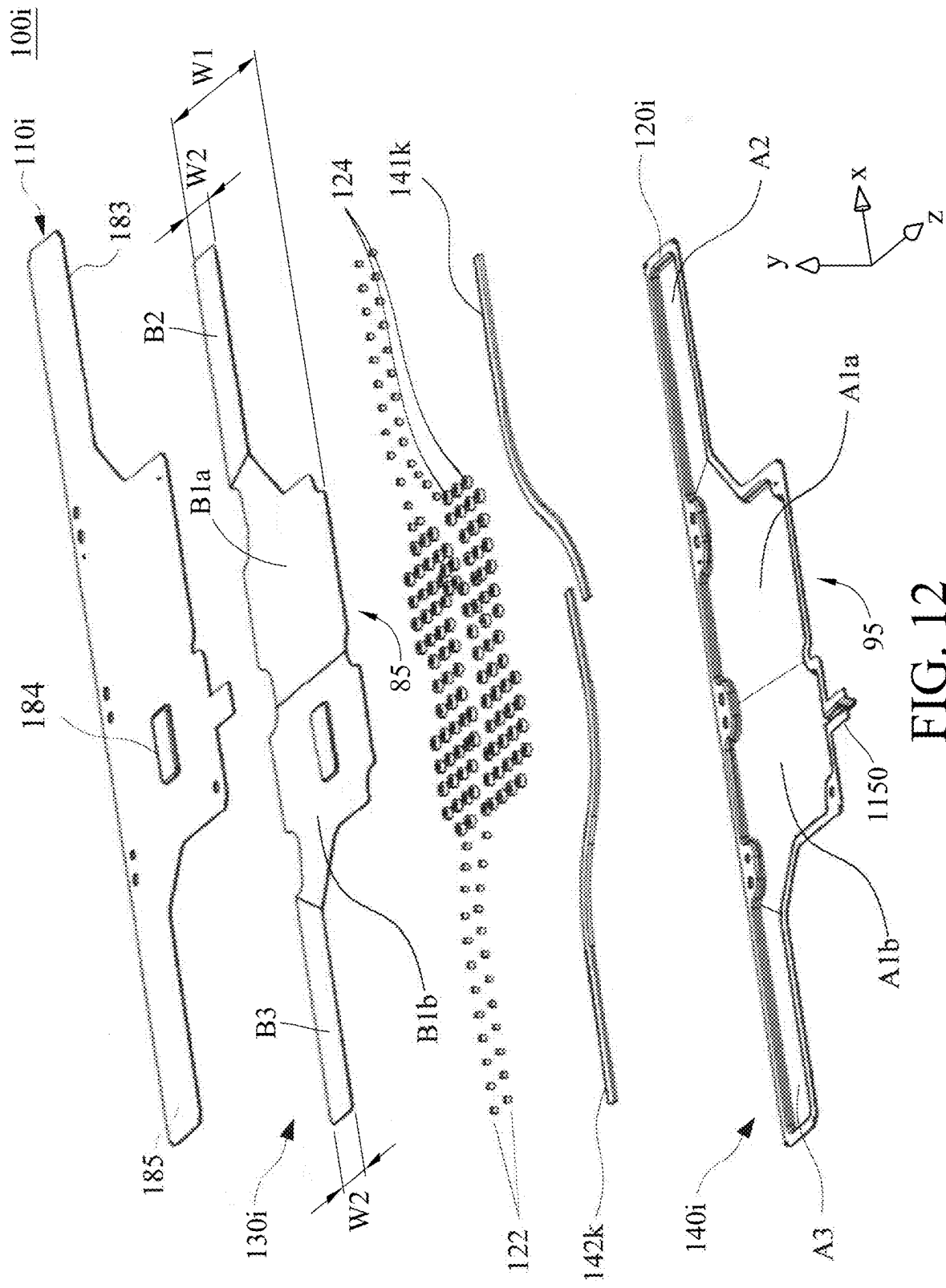


FIG. 11



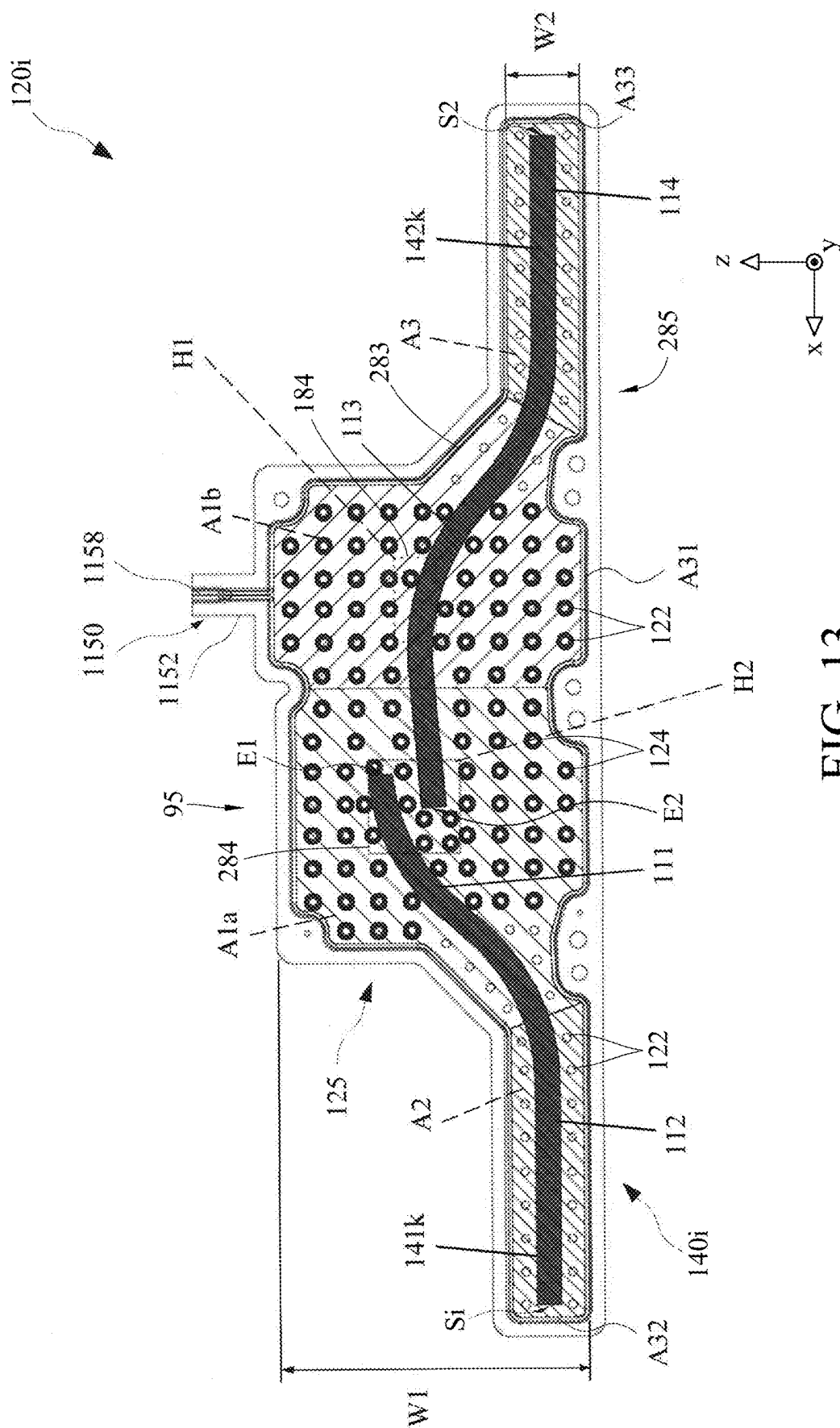


FIG. 13

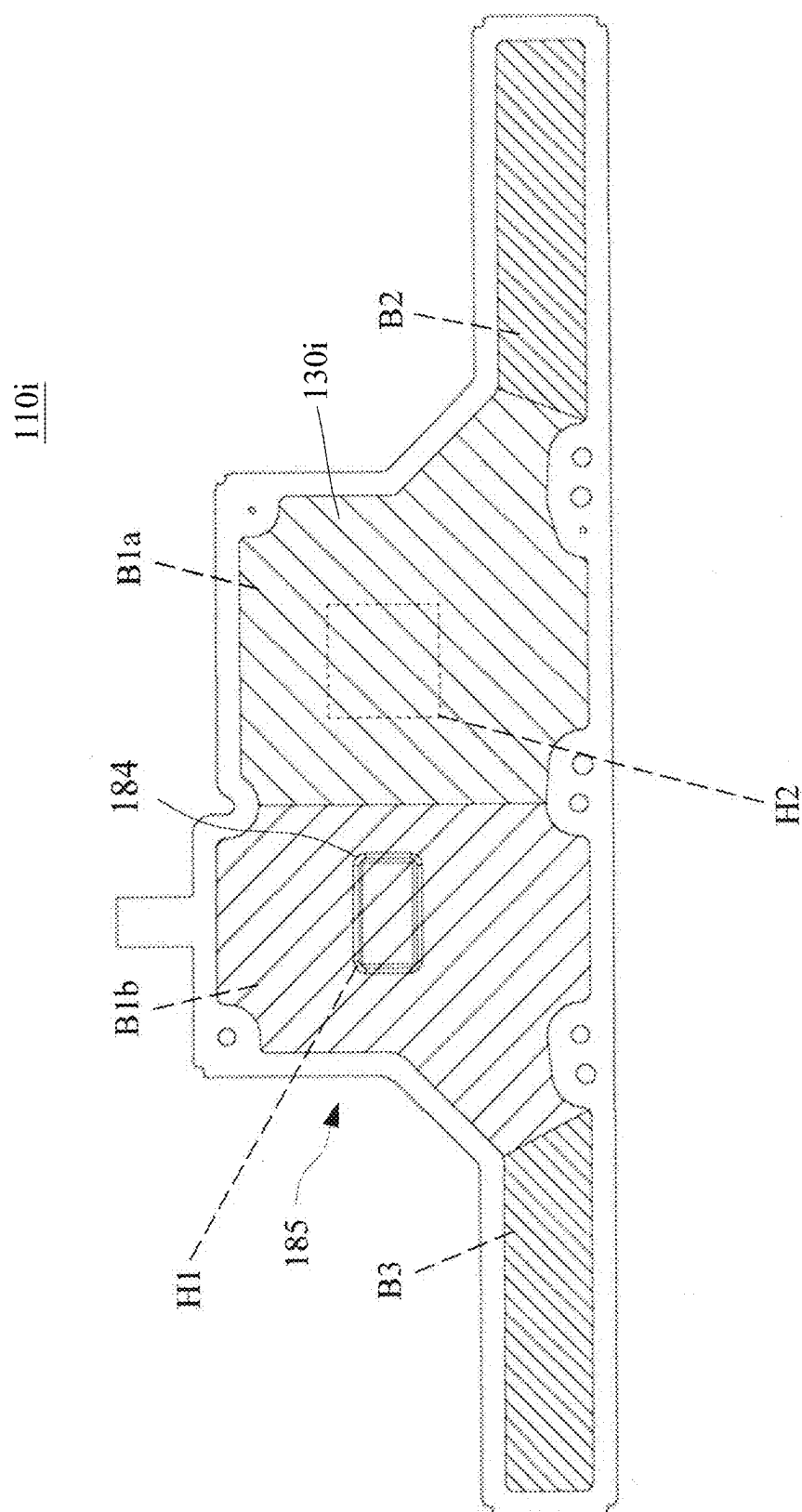


FIG. 14

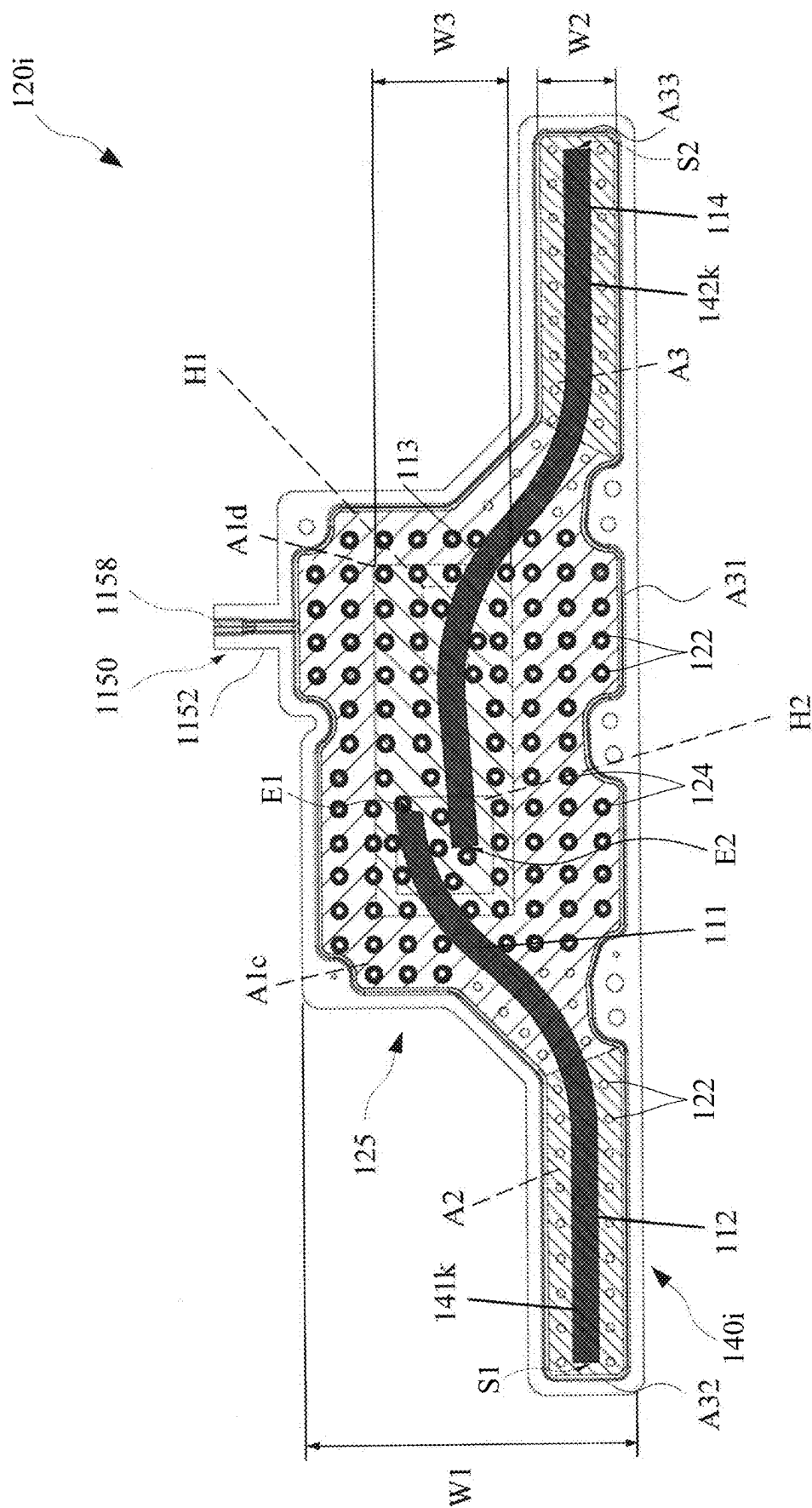


FIG. 15

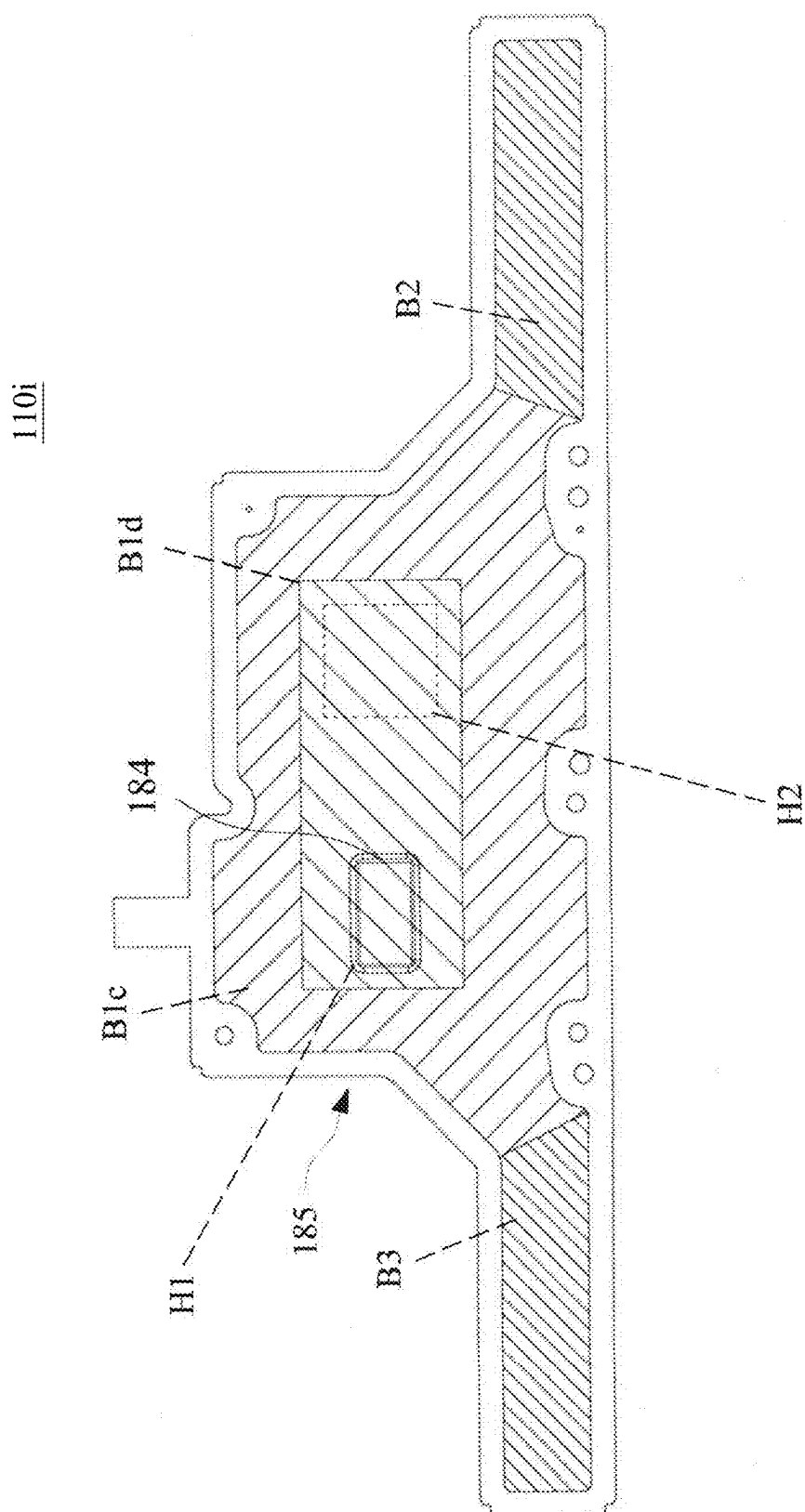


FIG. 16

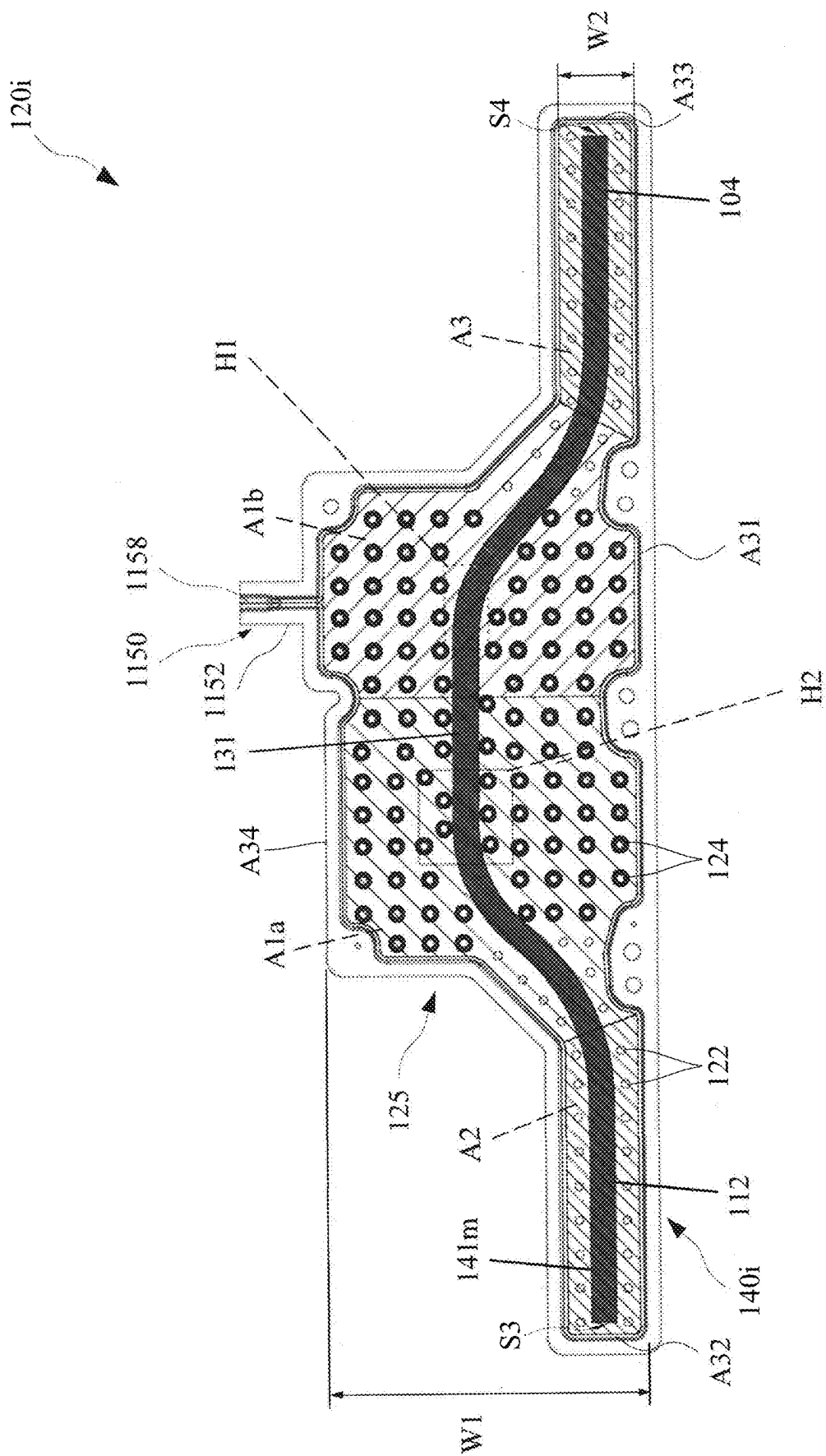


FIG. 17

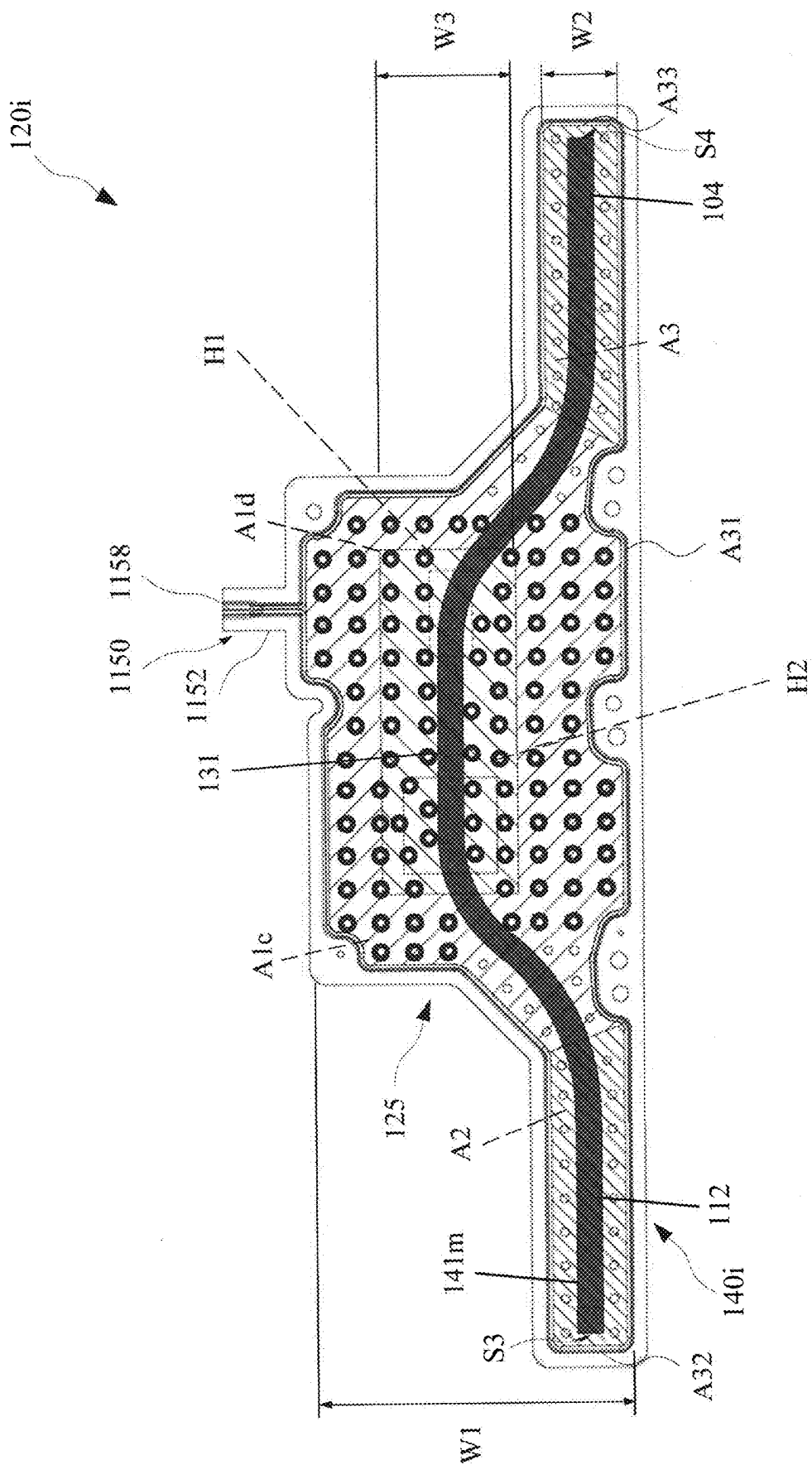


FIG. 18

HEAT DISSIPATION DEVICE HAVING IRREGULAR SHAPE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional application of U.S. application Ser. No. 17/665,440, filed on Feb. 4, 2022, which is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 16/711,233 filed Dec. 11, 2019, which in turn claims priority under 35 U.S.C. § 119 to U.S. provisional patent application Nos. 62/783,717 filed Dec. 21, 2018, and 62/798,480 filed Jan. 30, 2019. This application also claims priority under 35 U.S.C. § 119 to Chinese patent application no. CN202120353801.1 filed Feb. 8, 2021, in the China National Intellectual Property Administration (CNIPA). The entire contents of all these applications are incorporated herein by reference.

BACKGROUND

[0002] Vapor chambers have higher efficiency in heat dissipation than heat pipes. A vapor chamber includes a casing and a wick structure. The casing defines a chamber for accommodating cooling fluid. The wick structure is disposed in the chamber. The casing has an evaporation section for absorbing heat and a condensation section for dissipating heat. The cooling fluid is evaporated into gaseous state in the evaporation section, and then turns into liquid state in the condensation section and is carried back to the section area by the wick structure, thereby creating circulating cooling fluid.

[0003] Electronic products have become lighter, slimmer, and more compact, and, as a result, vapor chambers are required to have an irregular shape in order to not interfere or obstruct nearby electrical components. In such a case, portions of the vapor chamber are required to have reduced dimensions, and the cross-sectional area of the wick structure in these portions is reduced compared to its cross-sectional area in other portions of the vapor chamber. In a vertically orientated vapor chamber, when a heat source is in thermal contact with the upper portion of the vapor chamber, cooling fluid in the vapor chamber is required to circulated against gravity. However, the flow of the cooling fluid stagnates in the reduced dimension portions due to the smaller cross-sectional area of wick structure in these portions. Thus, the flow of the cooling liquid is restricted through the narrow area and circulation of the cooling fluid in the vapor chamber is inhibited, thereby affecting the operation of the vapor chamber.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is an exploded view of a vapor chamber according to embodiments of the disclosure.

[0006] FIG. 2A is a plan view of a second casing of the heat dissipation device in FIG. 1, according to embodiments of the disclosure.

[0007] FIG. 2B illustrates the encircled region 2B of FIG. 2A in greater detail.

[0008] FIG. 2C illustrates the encircled region 2C of FIG. 2A in greater detail.

[0009] FIG. 2D is a plan view of the second casing, according to embodiments of the disclosure.

[0010] FIG. 3 illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0011] FIG. 4 illustrates a plan view of a second casing including three wick structures, according to embodiments of the disclosure.

[0012] FIG. 5A illustrates a plan view of a second casing and a wick structure, according to embodiments of the disclosure.

[0013] FIG. 5B illustrates a plan view of the second casing of FIG. 5A including two wick structures, according to embodiments of the disclosure.

[0014] FIG. 6A illustrates a plan view of a second casing and a wick structure, according to some embodiments.

[0015] FIG. 6B illustrates the second casing of FIG. 6A including the wick structure of FIG. 6A and an additional wick structure, according to embodiments of the disclosure.

[0016] FIG. 7A illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0017] FIG. 7B illustrates a plan view of the second casing of FIG. 7A including a single wick structure, according to embodiments of the disclosure.

[0018] FIG. 8A illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0019] FIG. 8B illustrates a plan view of a second casing including a wick structure, according to some embodiments.

[0020] FIG. 8C illustrates a cross-sectional view of the second casing in FIG. 8B, according to embodiments of the disclosure.

[0021] FIG. 8D illustrates a plan view of the second casing including a single wick structure 180, according to embodiments of the disclosure.

[0022] FIG. 9 illustrates a plan view of a second casing including a wick structure, according to embodiments of the disclosure.

[0023] FIG. 10A is a cross-sectional view of a bundled wick structure including multiple wick fibers arranged around a central wick fiber, according to embodiments of the disclosure.

[0024] FIG. 10B-10F illustrate different configurations of the wick structures of FIG. 10A.

[0025] FIG. 10G is a cross-sectional view of a bundled wick structure including multiple wick fibers, according to embodiments of the disclosure.

[0026] FIGS. 10H, 10J, and 10K illustrate different configurations of the wick structures of FIG. 10G.

[0027] FIG. 11 illustrates a front view of a vapor chamber, according to embodiments of the disclosure.

[0028] FIG. 12 illustrates an exploded view of a vapor chamber in FIG. 11, according to embodiments of the disclosure.

[0029] FIG. 13 illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0030] FIG. 14 illustrates a plan view of a first casing including a sheet-like wick structure, according to embodiments of the disclosure.

[0031] FIG. 15 illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0032] FIG. 16 illustrates a plan view of a first casing including a sheet-like wick structure, according to embodiments of the disclosure.

[0033] FIG. 17 illustrates a plan view of a second casing including a wick structure, according to embodiments of the disclosure.

[0034] FIG. 18 illustrates a plan view of a second casing including a wick structure, according to embodiments of the disclosure.

DETAILED DESCRIPTION

[0035] It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific embodiments or examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, dimensions of elements are not limited to the disclosed range or values, but may depend upon process conditions and/or desired properties of the device. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for simplicity and clarity.

[0036] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “made of” may mean either “comprising” or “consisting of.”

[0037] FIG. 1 is an exploded view of a heat dissipation device 100 according to embodiments of the disclosure. FIG. 2A is a plan view of a second casing 120 of the heat dissipation device 100 in FIG. 1, according to embodiments of the disclosure. For the purposes of discussion herein, the heat dissipation device 100 is considered to be a vapor chamber. However, embodiments are not limited thereto and embodiments disclosed herein are equally applicable to other types of heat dissipation devices without departing from the scope of the disclosure.

[0038] Referring to FIGS. 1 and 2A, the vapor chamber 100 is a generally flat, planar structure including a first casing 110, a second casing 120, a sheet-like wick structure 130, and a wick structure 140. The vapor chamber 100 is referred to as a thin heat spreader. The vapor chamber 100 has a thickness less than approximately 1 millimeter.

[0039] The first casing 110 and the second casing 120 may be composed of, for example, oxygen-free copper, silicon-containing copper alloy, aluminum-containing copper alloy, a combination thereof, and the like. Referring to FIG. 2A, the second casing 120 includes a main body 121 and a plurality of supporting structures 122. The supporting structures 122 protrude (or otherwise project) from an inner surface 123 of the main body 121 towards the first casing 110 and contact the first casing 110. The supporting structures 122 reduce buckling of the first casing 110 and the second casing 120 and this limit deformation of the vapor chamber 100. In some embodiments, the supporting structures 122 are column or pillar shaped having a diameter of about 0.4 mm to about 0.8 mm. However, embodiments are not limited in this regard. Other shapes and sizes are also possible provided the plurality of supporting structures 122 limit buckling of the first casing 110 and second casing 120.

[0040] FIG. 2B illustrates the encircled region 2B of FIG. 2A in greater detail. In some embodiments, and as illustrated, the supporting structures 122 are arranged in rows (or columns) with immediately adjacent rows offset from each other. For example, as illustrated in FIG. 2B, the supporting structures 122 in row 131 are offset from the supporting structures 122 in row 133. In some embodiments, the distance D1 between alternate rows of supporting structures 122 (e.g., rows 131 and 135) is twice the distance between adjacent rows of supporting structures 122 (e.g., rows 131 and rows 133). In some embodiments, the distance D1 is between 3.5 mm to 4.5 mm and the distance D2 is between 1.5 mm to 2.5 mm. However, embodiments are not limited in this regard, and the distances D1 and D2 can be varied as required by application and design provided the plurality of supporting structures 122 limit buckling of the first casing 110 and second casing 120.

[0041] FIG. 2C illustrates the encircled region 2C of FIG. 2A in greater detail. As illustrated, the inner surface 123 includes a first fixing structure 175 along the outer periphery of the second casing 120 and a second fixing structure 177 surrounding the first fixing structure 175 and along the outer periphery of the second casing 120. In some embodiments, the first fixing structure 175 and the second fixing structure 177 define a channel or groove 173 therebetween. The inner surface of the first casing 110 includes a corresponding protrusion that is received in the groove 173 for securing the first casing 110 and the second casing 120 to each other.

[0042] In some embodiments, the thickness of the first casing 110 is smaller than the thickness of the second casing 120. As an example, the thickness of first casing 110 is about 0.1 mm to about 0.15 mm and the thickness of the second casing 120 is about 0.2 mm to about 0.35 mm. An overall thickness of the vapor chamber 100 including the first casing 110 and the second casing 120 is about 0.3 mm to 0.5 about mm. However, embodiments are not limited in this regard, and the thickness of the first casing 110 may be greater than the thickness of the second casing 120.

[0043] The inner surface 123 is opposite to the outer surface 125 of the main body 121. The outer surface 125 forms part of the outer surface of the vapor chamber 100 and is substantially planar. The first casing 110 is connected to the main body 121 of the second casing 120 by welding, soldering, brazing, or diffusion bonding. The second casing 120 has a first or “broad” portion A1 and a second or “narrow” portion A2. The first portion A1 has a width W1 and the second portion A2 has a width W2. The width W1

is larger than the width W2. The first portion has length L1 and the second portion has a length L2. The sum of the lengths L1 and L2 is the length L3 of the longest side of the vapor chamber 100. It will thus be understood that each of the first portion A1 and the second portion A2 has a different cross-sectional area. In an embodiment, the width W1 of the first portion A1 is approximately 80 millimeters (mm), the width W2 of the second portion A2 is approximately 18 mm. The length L2 is about 90 mm to about 100 mm, and the length L3 is about 120 mm to about 130 mm. However, embodiments are not limited in this regard and can be varied depending on the application and design requirements. In other embodiments, the width W2 of the second portion A2 may be equal to or less than half of the width W1 of the first portion A1. In still other embodiments, the width W2 of the second portion A2 may be equal to or less than one-third of the width W1 of the first portion A1.

[0044] The wick structure 130 is disposed between the first casing 110 and the second casing 120. The wick structure 130 includes, for example, copper mesh. The wick structure 130 contacts the first casing 110 and the wick structure 140. The wick structure 130 provides a flow path for the cooling fluid circulating in the vapor chamber 100.

[0045] The wick structure 140 may be or include a bundle of copper wires twisted to form a single helical structure. The wick structure 140 is located in the gaps between the supporting structures 122 and contacts the inner surface 123 of the second casing 120, and thereby is in contact with the second casing 120. Although the wick structure 140 has been disclosed as a bundle of wires twisted into a helical structure, embodiments are not limited thereto. In other embodiments, the wick structure 140 may be or include a copper sintered powder wick structure including copper wires interlaced or twisted into a bundle. In some other embodiments, the wick structure 140 may be or include a plurality of the bundles of wires being interlaced or twisted. In still other embodiments, the wick structure 140 includes a screen mesh wick structure or groove wick structure.

[0046] In an embodiment and as illustrated, the wick structure 140 is longitudinally extending structure and relatively straight (without any curves or bends) disposed proximate upper ends of the first portion A1 and second portion A2. One end S of the wick structure 140 is located in the second portion A2 and proximate (but not contacting) a lateral edge A21 of the second portion A2. The lateral edge A21 is outer edge of the second portion A2, and thereby of the second casing 120. In some embodiments, the end S of the wick structure 140 may not contact with the edge A21 of the second portion A2, and may be spaced apart from the edge A21 of the second portion A2. The other longitudinally opposite end E of the wick structure 140 is located in the first portion A1. A heat source H is attached to the outer surface of the first casing 110 and thereby the outer surface of the vapor chamber 100 using a thermal paste, or other known techniques. FIG. 2 illustrates a vertical projection (dashed box) of the heat source H on the second casing 120. The heat source H is illustrated as square shaped for the sake of illustration. It will be understood that the shape of the heat source H is not limited to a square shape and the heat source H can have any desired shaped. The end E of the wick structure 140 is spaced (e.g., vertically, in FIG. 2) from the heat source H by a distance D. The distance D is approximately 18.5 mm, but embodiments are not limited thereto. In some embodiments, the end E of the wick structure 140 may

overlap the heat source H or the end E and boundary (edges) of the heat source H may be coincident. It will be understood that the distance D between the end E of the wick structure 140 and the heat source H is not limited to any particular value and may be adjusted according to the size or shape of the vapor chamber and as per user and design requirements.

[0047] The vapor chamber 100 includes a working appendage 1150 that is cooperatively formed by a protrusion 1152 of the second casing 120 and a corresponding protrusion of the first casing 110. The working appendage 1150 includes a charging channel 1158 formed by the first casing 110 and the second casing 120. The charging channel 1158 fluidly communicates with the interior cavity of the vapor chamber 100 formed by the inner surface 123 of the second casing 120 and the inner surface of the first casing 110. The working appendage 1150 is used to fill the interior cavity of the vapor chamber 100 with working fluid and vacuuming out the air from the interior cavity. In some embodiments, the working appendage 1150 is centrally located along the width W1. However, in other embodiments, the working appendage 1150 is located offset from the central location or located on other sides or surfaces of the vapor chamber 100.

[0048] FIG. 2D is a plan view of the second casing 120, according to embodiments of the disclosure. In contrast to FIG. 2A, as illustrated, the wick structure 140 is located proximate a lower end of the second portion A2. The distance D between the end E of the wick structure 140 and the heat source H is thus reduced. The wick structure 140 is spaced from the lateral edge A21 of the second portion A2.

[0049] In some embodiments, the vapor chamber 100 may include more than one wick structure 140. FIG. 3 illustrates a plan view of a second casing 120a including two wick structures 141a and 142a, according to embodiments of the disclosure. The second casing 120a may be used instead of the second casing 120 in FIGS. 1 and 2 in the vapor chamber 100. The second casing 120a may be similar in some respects to the second casing 120 in FIG. 2, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0050] As illustrated, each of the wick structures 141a and 142a are longitudinally extending structures and are straight structures (without bends or curves) and are disposed on the second casing 120a. The wick structures 141a and 142a are located in the gaps between the supporting structures 122 and each contacts the inner surface 123 of the second casing 120a. In an embodiment, the wick structures 141a and 142a have different lengths. For example, and as illustrated, the wick structure 141a is longer than the wick structure 142a. The wick structures 141a and 142a contact each other along the longitudinal edges thereof and are arranged side by side. One end Sa1 of the wick structure 141a is located in the second portion A2 and is in contact with the edge A21 of the second portion A2. The longitudinally opposite end Ea1 of the wick structure 141a is located in the first portion A1 and spaced (e.g., vertically, in FIG. 3) from the heat source H. One end Sa2 of the wick structure 142a is located in the second portion A2, and the longitudinally opposite end Ea2 of the wick structure 142a is located in the first portion A1 and spaced (e.g., vertically, in FIG. 3) from the heat source H. As illustrated, the ends Ea1 and Ea2 are coincident with each other (e.g., located at a same distance from the edge A21). However, in other embodiments, the ends Ea1 and

Ea2 are non-coincident (e.g., located at different distances from edge A21) in the first portion A1.

[0051] Compared to the embodiment in FIG. 2, the second casing 120a in FIG. 3 includes the additional wick structure 142a extending from the second portion A2 toward the heat source H. The wick structure 142a improves circulation of cooling fluid in the vapor chamber 100, for example, against gravity. Thus, the efficiency of the vapor chamber 100 in dissipating heat is improved.

[0052] FIG. 4 illustrates a plan view of a second casing 120b including three wick structures 141a, 142a, and 143a, according to embodiments of the disclosure. The second casing 120b may be used instead of the second casing 120 in FIGS. 1 and 2 in the vapor chamber 100. The second casing 120b may be similar in some respects to the second casings 120 and 120a in FIGS. 2 and 3, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0053] Referring to FIG. 4, each of the wick structures 141a, 142a, and 143a are longitudinally extending structures and are straight structures (without bends or curves) and disposed on the second casing 120b. The wick structures 141a, 142a, and 143a are located in the gaps between the supporting structures 122 and each contacts the inner surface 123 of the second casing 120b. In an embodiment, the wick structures 141a, 142a, and 143a have different lengths. For example, and as illustrated, the wick structure 141a is longer than the wick structures 142a and 143a. The wick structures 141a and 142a contact each other along the longitudinal edges thereof and arranged side by side. The wick structure 143a is spaced (e.g., vertically, in FIG. 4) from the wick structures 141a and 142a. One end Sa1 of the wick structure 141a is located in the second portion A2 and contacts the edge A21 of the second portion A2. The longitudinally opposite end Ea1 of the wick structure 141a is located in the first portion A1 and spaced from the heat source H (illustrated in phantom). One end Sa2 of the wick structure 142a is located in the second portion A2, and the longitudinally opposite end Ea2 of the wick structure 142a is located in the first portion A1 and spaced from the heat source H. As illustrated, the ends Ea1 and Ea2 are coincident with each other (e.g., located at a same distance from the edge A21). However, in other embodiments, the ends Ea1 and Ea2 are non-coincident. One end Ea3 of the wick structure 143a is collinear (e.g., aligned) with the heat source H on the outer surface of first casing 110, and the other end Sa3 of the wick structure 143a is located in the first portion A1 and contacts a vertical edge A11 of the first portion A1. As illustrated, the wick structure 143a is entirely within the first portion A1. The horizontal distance between the end Ea3 of the wick structure 143a and the heat source H is less than the vertical distance between the end Ea2 of the wick structure 142a and the heat source H. Although, the wick structures 141a, 142a, and 143a are indicated as having different lengths, embodiments are not limited thereto. In some embodiments, the wick structures 142a and 143a may have the same lengths. In other embodiments, the end Sa1 of the wick structure 141a and end Sa2 of the wick structure 142a may be located at a same distance from the edge A21.

[0054] Compared to embodiment of FIG. 3, the second casing 120b in FIG. 4 includes an additional wick structure 143a extending from the edge of the first portion A1 toward the heat source H. The wick structure 143a further improves circulation of cooling fluid in the vapor

chamber 100, for example, against gravity. Thus, the efficiency of the vapor chamber 100 in dissipating heat is further improved.

[0055] FIG. 5A illustrates a plan view of a second casing 120c and a wick structure 140c, according to embodiments of the disclosure. The second casing 120c may be used instead of the second casing 120 in FIGS. 1 and 2 in the vapor chamber 100. The second casing 120c may be similar in some respects to the second casing 120 in FIG. 2, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0056] As illustrated, the wick structure 140c is disposed in the second casing 120c and has at least one bend. The wick structure 140c is located in the gaps between the supporting structures 122 and contacts the inner surface 123 of the second casing 120. An end Sc of the wick structure 140c is located in the second portion A2 and contacts the edge A21 of the second portion A2 and the opposite end Ec of the wick structure 140c is located in the first portion A1 and overlaps at least part of the heat source H (illustrated in phantom) attached to the outer surface of the first casing 110. In an example, and as illustrated the wick structure 140c at least partially overlaps one side or end of the heat source H that is closer to the wick structure 140c. For the sake of explanation, it is assumed that the heat source H has a generally rectangular shape including a first side L1 and a second side L2 opposite the first side L1, and a third side L3 and a fourth side L4 opposite the third side L3. The first side L1 is closer to the edge A21. The fourth side L4 is closer to the wick structure 140c. The wick structure 140c overlaps the fourth side L4 of the heat source H.

[0057] As illustrated, the wick structure 140c includes two sections 1401 and 1402, having lengths Y1 and Y2, respectively. The sections 1401 and 1402 are connected to each other at an angle (greater than 0° and less than 180°). The overall length of the wick structure 140c is the sum of the lengths Y1 and Y2. In an example, the overall length is approximately 118 mm, but the disclosure is not limited thereto. In other embodiments, the overall length of the wick structure 140c increased or decreased as long as the wick structure 140c is accommodated within the vapor chamber 100. In an embodiment, an extent of the wick structure 140c may be equal to the minimum distance X between the end Sc of the wick structure 140c and the heat source H. More specifically, the minimum distance X is measured from the end Sc to a point on the heat source H nearest to the end Sc as projected vertically on the inner surface 123. For instance, the distance X is approximately 105 mm. The heat dissipation efficiency of the vapor chamber including the wick structure 140c is relatively higher when the overall length of the wick structure 140c is equal to or less than about 126 mm.

[0058] The wick structure 140c decreases a distance the cooling fluid has to circulate in the vapor chamber 100. The length of the wick structure 140c is not limited to any particular length and may be adjusted according to user and design requirements.

[0059] FIG. 5B illustrates a plan view of the second casing 120c including the wick structure 140c and a wick structure 150c, according to embodiments of the disclosure. The wick structure 150c is shaped and sized (or otherwise configured) similar to the wick structure 140c. The wick structure 150c contacts the wick structure 140c at a bottom side thereof. As

illustrated, the wick structure **150c** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120c**. An end **155** of the wick structure **150c** is located in the second portion **A2** and contacts the edge **A21** of the second portion **A2** and the opposite end **157** of the wick structure **150c** is located in the first portion **A1** and overlaps at least part of the heat source **H** (illustrated in phantom) attached to the outer surface of the first casing **110**. As depicted, the ends **155** and **157** of the wick structure **150c** are aligned with the ends **Sc** and **Ec** of the wick structure **140c**.

[0060] In some embodiments, the wick structure **150c** overlaps a same side of the heat source **H** as overlapped by the wick structure **140c**. In other embodiments, the wick structure **150c** overlaps a different side of the heat source **H**. The extent of the wick structure **150c** is the same as the extent of the wick structure **140c**.

[0061] The wick structure **150c** includes sections **1501** and **1502** having lengths **Y1** and **Y2**, respectively, and connected to each other at a same angle (greater than 0° and less than 180°) as the sections **1401** and **1402** of the wick structure **140c**. However, in other embodiments, the lengths of the sections **1501** and **1502** are different from the lengths of the sections **1401** and **1402**. In some embodiments, the length of the section **1501** is smaller than the length of the section **1401**. In some other embodiments, the length of the section **1502** is longer than the length of the section **1402**.

[0062] The location of the wick structure is not restricted to any particular location in the vapor chamber. FIG. 6A illustrates a plan view of a second casing **120d** and a wick structure **140d**, according to some embodiments. The second casing **120d** may be used instead of the second casing **120** of FIGS. 1 and 2 in the vapor chamber **100**. The second casing **120d** may be similar in some respects to the second casing **120c** in FIG. 5A, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0063] Referring to FIG. 6A, the wick structure **140d** has at least one bend and is disposed on the second casing **120d**. The wick structure **140d** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120d**. One end **Sd** of the wick structure **140d** is located in the second portion **A2**, and the other opposite end **Ed** of the wick structure **140d** is located in the first portion **A1** and overlaps the heat source **H** attached to the outer surface of the first casing **110**. Assuming, for the sake of explanation, that the heat source **H** is the same as the heat source **H** in FIG. 5A, the wick structure **140d** overlaps the first side **L1**. As illustrated, the section having length **Y1** of the wick structure **140d** contacts a lower horizontal edge **A22** of the second portion **A2**.

[0064] The wick structure **140d** has a relatively shorter length than the wick structure **140c** of FIG. 5A. The overall length of the wick structure **140d** is approximately 108 mm, but the disclosure is not limited thereto. In other embodiments, the overall length of the wick structure **140d** increased or decreased as long as the wick structure **140c** is accommodated within the vapor chamber **100**. An extent of the wick structure **140d** may be equal to the minimum distance **X** between the end **Sd** of the wick structure **140d** and the heat source **H**. More specifically, the minimum distance **X** is measured from the end **Sd** to a point on the heat source **H** nearest to the end **Sd** projected vertically on the inner surface **123**. For example, the distance **X** is approxi-

mately 100 mm. The heat dissipation efficiency of the vapor chamber including the wick structure **140c** is relatively higher when the overall length of the wick structure **140d** is equal to or less than about 120 mm.

[0065] FIG. 6B illustrates the second casing **120d** including the wick structure **140d** and a wick structure **150d**, according to embodiments of the disclosure. The wick structure **150d** is shaped and sized (or otherwise configured) similar to the wick structure **140d**. The wick structure **150d** contacts the wick structure **140d** at the top side thereof. The wick structure **150d** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120d**.

[0066] An end **165** of the wick structure **150d** is located in the second portion **A2** and is spaced from the edge **A21** of the second portion **A2** and the opposite end **167** of the wick structure **150d** is located in the first portion **A1** and overlaps at least part of the heat source **H** (illustrated in phantom) attached to the outer surface of the first casing **110**. As depicted, the ends **165** and **167** of the wick structure **150d** are aligned with the ends **Sd** and **Ed** of the wick structure **140d**.

[0067] In some embodiments, the wick structure **150d** overlaps a same side of the heat source **H** as overlapped by the wick structure **140d**. In other embodiments, the wick structure **150d** overlaps a different side of the heat source **H**. The extent of the wick structure **150d** is the same as the extent of the wick structure **140d**.

[0068] The wick structure **150d** sections **1501** and **1502** having lengths **Y1** and **Y2**, respectively, and connected to each other at a same angle (greater than 0° and less than 180°) as the sections **1401** and **1402** of the wick structure **140d**. However, in other embodiments, the lengths of the sections **1501** and **1502** are different from the lengths of the sections **1401** and **1402**. In some embodiments, the length of the section **1501** is smaller than the length of the section **1401**. In some other embodiments, the length of the section **1502** is longer than the length of the section **1402**.

[0069] Referring to FIGS. 5A and 6, the wick structure **140d** in FIG. 6 is shorter than the wick structure **140c** in FIG. 5A, and this reduces the path for the cooling liquid to circulate in the vapor chamber **100**. The length of the wick structure **140d** is not limited to any particular length and may be adjusted according to user and design requirements.

[0070] According to some embodiments, the shapes of the first casing and the second casing are not limited to any particular shape. FIG. 7A illustrates a plan view of a second casing **120e** including two wick structures **141e** and **142e**, according to embodiments of the disclosure. The second casing **120e** may be used in a vapor chamber that includes a first casing and a sheet-like wick structure similar to the first casing **110** and the sheet-like wick structure **130** in FIG. 1, respectively, but each having a shape corresponding to the shape of the second casing **120e**.

[0071] As illustrated in FIG. 7A, the second casing **120e** is generally H-shaped and has a first portion **A1**, a second portion **A2**, and a third portion **A3**. The third portion **A3** is located between the first and second sections **A1** and **A2**. The first portion **A1** has a width **W1**, the second portion **A2** has a width **W2**, and the third portion **A3** has a width **W3**. The width **W1** is equal to the width **W2**, and the width **W3** is smaller than the width **W1**. However, in other embodiments the widths **W1** and **W2** may be different, but each greater than the width **W3**. The wick structures **141e** and **142e** are

disposed on the second casing 120e. The wick structures 141e and 142e each have a generally L-shape and are located in the gaps between the supporting structures 122 and each contacts the inner surface 123 of the second casing 120e.

[0072] The wick structure 141e includes a first section 151 that is disposed in the first portion A1, second portion A2, and third portion A3. The wick structure 141e includes a second section 152 disposed in the third portion A3 and at an angle relative to the first section 151. The first section 151 is connected to the second section 152 in the third portion A3. In an example, the first section 151 is perpendicular to the second section 152. However, embodiments are not limited thereto, and the first section 151 is connected to the second section 152 at angles greater than or less than 90°.

[0073] The wick structure 142e includes a first section 153 that is disposed in the first portion A1, second portion A2, and third portion A3. The wick structure 142e includes a second section 154 disposed in the third portion A3 and at an angle relative to the first section 153. The first section 153 is connected to the second section 154 in the third portion A3. In an example, the first section 153 is perpendicular to the second section 154. However, embodiments are not limited thereto, and the first section 153 is connected to the second section 154 at angles greater than or less than 90°. The first section 151 of the wick structure 141e and the first section 153 of the wick structure 142e contact each other.

[0074] The first section 151 includes an end Ee1 of the wick structure 141e in the first portion A1. The end Ee1 overlaps the heat source H attached to the outer surface of the first casing 110. The first section 153 includes an end Ee2 of the wick structure 142e in the first portion A1. The end Ee2 overlaps the heat source H attached to the outer surface of the first casing 110.

[0075] The second section 152 includes an end Se1 of the wick structure 141e in the third portion A3. The second section 154 includes an end Se2 of the wick structure 142e in the third portion A3.

[0076] As illustrated, the end Se1 of the wick structure 141e is located proximate (non-contacting) an upper edge A22 of the second portion A2 and the end Se2 of the wick structure 142e is located proximate a lower edge A23 of the second portion A2 opposite the upper edge A22. The second sections 152 and 154 are disposed proximate the edge A21 of the second portion A2. The edge A21 connects the upper edge A22 and lower edge A23. In other embodiments, the end Se1 may contact the upper edge A22, the end Se2 may contact the lower edge A23, and the second sections 152 and 154 contact the edge A21.

[0077] Instead of two wick structures 141e and 142e, some embodiments include a single wick structure. FIG. 7B illustrates a plan view of the second casing 120e including a single wick structure 170, according to embodiments of the disclosure. The wick structure 170 includes a first section 171 disposed in the first portion A1, second portion A2, and third portion A3. The wick structure 170 includes a second section 172 disposed in the third portion A3 and at an angle relative to the first section 171. The first section 171 is connected to the second section 172 in the third portion A3. In an example, the first section 171 is perpendicular to the second section 172. However, embodiments are not limited thereto, and the first section 171 is connected to the second section 172 at angles greater than 0° or less than 90°.

[0078] The first section 171 includes an end Ee1 of the wick structure 170 in the first portion A1. The end Ee1

overlaps the heat source H attached to the outer surface of the first casing 110. The second section 172 includes an end Se1 of the wick structure 170 in the third portion A3. The second section 172 includes an end Se2 opposite end Se1 located in the third portion A3. The second section 172 is disposed proximate the edge A21 of the second portion A2. [0079] FIG. 8A illustrates a plan view of a second casing 120f including two wick structures 141f and 142f, according to embodiments of the disclosure. The second casing 120f may be used in a vapor chamber that includes a first casing and a sheet-like wick structure similar to the first casing 110 and the sheet-like wick structure 130 in FIG. 1, respectively, but each having a shape corresponding to the shape of the second casing 120f.

[0080] The second casing 120f includes a first portion A1 disposed between a second portion A2 and a third portion A3. The first portion A1 has a width W1, and the second portion A2 and the third portion A3 each have a width W2. The width W1 is larger than the width W2. In other embodiments, the second portion A2 and the third portion A3 may have different widths, but smaller than the width W1. The wick structures 141f and 142f are disposed on the second casing 120f. The wick structures 141f and 142f each have a generally L-shape and are located in the gaps between the supporting structures 122 and each contacts the inner surface 123 of the second casing 120f.

[0081] The wick structure 141f includes a first section 161 that is disposed in the first portion A1. The wick structure 141f includes a second section 162 disposed in the first portion A1 and the second portion A2 and at an angle relative to the first section 161. The first section 161 is connected to the second section 162 in the first portion A1. In an example, the first section 161 is perpendicular to the second section 162. However, embodiments are not limited thereto, and the first section 161 is connected to the second section 162 at angles greater than or less than 90°.

[0082] The wick structure 142f includes a first section 163 that is disposed in the first portion A1. The wick structure 142f includes a second section 164 disposed in the third portion A3 and at an angle relative to the first section 163. The first section 163 is connected to the second section 164 in the first portion A1. In an example, the first section 163 is perpendicular to the second section 164. However, embodiments are not limited thereto, and the first section 163 is connected to the second section 164 at angles greater than or less than 90°. The first section 161 of the wick structure 141f and the first section 163 of the wick structure 142f contact each other.

[0083] The first section 161 includes an end Ef1 of the wick structure 141f located in the first portion A1. The end Ef1 overlaps the heat source H attached to the outer surface of the first casing 110. The first section 163 includes an end Ef2 of the wick structure 142f in the first portion A1. The end Ef2 overlaps the heat source H attached to the outer surface of the first casing 110.

[0084] The second section 162 includes an end Sf1 of the wick structure 141f in the second portion A2. The second section 164 includes an end Sf2 of the wick structure 142f in the third portion A3.

[0085] As illustrated, the end Sf1 of the wick structure 141f is located proximate an edge A32 of the second portion A2 and the end Sf2 of the wick structure 142f is located proximate (non-contacting) an edge A33 of the third portion A3 opposite the edge A32. The second sections 162 and 164

are disposed proximate (but not contacting) an edge **A31** of the second casing **120f** that forms (or otherwise defines) the first portion **A1**, the second portion **A2**, and the third portion **A3** of the second casing **120f**. The edge **A31** connects the edge **A32** and edge **A33**. In other embodiments, the end **Sf1** may contact the edge **A32**, the end **Sf2** may contact the edge **A33**, and the second sections **162** and **164** contact the edge **A31**.

[0086] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the heat source **H** is located at a level higher than the second portion **A2** and third portion **A3**), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the heat source **H**. The cooling fluid (in vapor state) flows toward the second portion **A2** and the third portion **A3** and turns back to liquid state, and the cooling fluid can flow to the first portion **A1** via the wick structures **141f** and **142f**. This configuration decreases a temperature difference between the first portion **A1** and the second portion **A2** and the third portion **A3** by around 4 to 15 degrees Celsius. The cooling fluid continuously circulates between the first portion **A1** and the second portion **A2** and the third portion **A3**, thereby dissipating the heat generated by the heat source **H**.

[0087] FIG. **8B** illustrates a plan view of a second casing **120g** including a wick structure **142f**, according to some embodiments. FIG. **8C** illustrates a cross-sectional view of the second casing **120g** taken along line **8B-8B** in FIG. **8B**. The second casing **120g** may be similar in some respects to the second casing **120f** in FIG. **8A**, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0088] As illustrated in FIG. **8B**, the second casing **120g** includes a single wick structure **142f**. The wick structure **141f** is absent. The wick structure **142f** is spaced from the edge **A31**. In some embodiments, the second section **164** is disposed mid-way in the third portion **A3**. In some embodiments, and as illustrated, the first section **163** of the wick structure **142f** is centrally located in the first portion **A1**. However, embodiments are not limited in this regard and in some other embodiments, the first section **163** is located offset from the center of the first portion **A1**.

[0089] In some embodiments, the second casing **120g** includes the wick structure **141f** (FIG. **8A**) and the wick structure **142f** is absent. The wick structure **141f** is spaced from the edge **A31**. In some embodiments, the second section **162** of the wick structure **141f** is disposed mid-way in the second portion **A2**. In some embodiments, the first section **161** (FIG. **8A**) of the wick structure **141f** is centrally located in the first portion **A1**. However, in some other embodiments, the first section **161** is located offset from the center of the first portion **A1**.

[0090] Instead of two wick structures **141f** and **142f**, some embodiments include a single wick structure. FIG. **8D** illustrates a plan view of the second casing **120f** including a single wick structure **180**, according to embodiments of the disclosure. The wick structure **180** includes a first section **181** disposed in the first portion **A1**. The wick structure **180** includes a second section **182** disposed in the first portion **A1**, second portion **A2**, and third portion **A3** and at an angle relative to the first section **181**. The first section **181** is connected to the second section **182** in the first portion **A1**. In an example, the first section **181** is perpendicular to the second section **182**. However, embodiments are not limited

thereto, and the first section **181** is connected to the second section **182** at angles greater than 0° or less than 90° .

[0091] The first section **181** includes an end **Ef1** of the wick structure **180** in the first portion **A1**. The end **Ef1** overlaps the heat source **H** attached to the outer surface of the first casing **110**. The second section **182** includes an end **Sf1** of the wick structure **180** in the second portion **A2** and an end **Sf2** opposite end **Sf1** located in the third portion **A3**. The second section **182** is disposed proximate (but not contacting) the edge **A31** that forms (or otherwise defines) the first portion **A1**, the second portion **A2**, and the third portion **A3** of the second casing **120f**. Although not illustrated, a working appendage similar to the working appendage **1150** in FIG. **2A** is included in the second casing **120f** in FIGS. **8A**, **8B**, and **8D**. The working appendage is located at any desirable location along the sides of the second casing **120f**.

[0092] The shape of the wick structure is not limited to any particular shape. In some embodiments, the wick structure may be in a zig-zag shape or any other desired shape.

[0093] FIG. **9** illustrates a plan view of a second casing **120h** including a wick structure **141h**, according to embodiments of the disclosure. The second casing **120h** may be used in a vapor chamber that includes a first casing and a sheet-like wick structure similar to the first casing **110** and the sheet-like wick structure **130** in FIG. **1**, respectively, but each having a shape corresponding to the shape of the second casing **120h**.

[0094] As illustrated in FIG. **9**, the second casing **120h** is generally U-shaped and has a horizontally orientated first portion **A1**, and a second portion **A2** and a third portion **A3**, each vertically oriented. The first portion **A1** is located between the second and third sections **A2** and **A3**. The first portion **A1** has a width **W1**, the second portion **A2** has a width **W2**, and the third portion **A3** has a width **W3**. The width **W2** is equal to the width **W3**, and the width **W1** is less than the widths **W2** and **W3**. However, in other embodiments the widths **W2** and **W3** may be different, but each greater than the width **W1**. The wick structures **141h** is disposed in the second casing **120h** and in the gaps between the supporting structures **122**. The wick structures **141h** contacts the inner surface **123** of the second casing **120h**.

[0095] The wick structure **141h** includes a first section **191** horizontally oriented and disposed in the first portion **A1**, second portion **A2**, and third portion **A3**. The wick structure **141h** includes two vertically oriented sections, a second section **192** disposed in the second portion **A2** and a third section **193** disposed in the third portion **A3**. Each of the first section **191**, second section **192**, and third section **193** are straight, longitudinal structures without any bends. The second section **192** and the third section **193** are angled relative to the first section **191**. The second section **192** and the third section **193** are connected to the first section **191** in the second portion **A2** and the third portion **A3**, respectively. In an embodiment, and as illustrated, the second section **192** and the third section **193** are perpendicular to the first section **191**. However, embodiments are not limited in this regard, and the second section **192** and the third section **193** are connected to the first section **191** at angles greater than 0° or less than 90° .

[0096] The first section **191** includes an end **Sg2** in the second portion **A2**, and proximate an outer vertical edge **A22** of the second portion **A2**. The first section **191** includes an end **Sg3** in the third portion **A3** longitudinally opposite the

end Sg2, and proximate an outer vertical edge A23 of the third portion A3. The first section 191 is located proximate a bottom edge A21 of the first portion A1, second portion A2, and third portion A3.

[0097] The second section 192 includes an end Eg2 in the second portion A2. The end Eg2 is longitudinally opposite the end of the second section 192 connected to the first section 191. Similarly, the third section 193 includes an end Eg3 in the third portion A3. The end Eg3 is longitudinally opposite the end of the third section 193 connected to the first section 191.

[0098] As illustrated, the end Eg2 overlaps the heat source H attached to the outer surface of the first casing 110 and over the second portion A2. However, in other embodiments, the end Eg3 overlaps the heat source H attached to the outer surface of the first casing 110 and over the third portion A3. A working appendage 1150 is located in the third portion A3. However in other embodiments, the working appendage 1150 is located in the first portion A1 or the second portion A2 along any desired edge.

[0099] FIG. 10A is a cross-sectional view of a bundled wick structure including multiple wick fibers 1010 arranged around a central wick fiber 1020. In an embodiment, the wick fibers 1010 and 1020 include copper. However, the wick fibers 1010 and/or 1020 can include other materials that facilitate the flow of condensed fluid by capillary force through the wick structure formed using the wick fibers 1010 and 1020.

[0100] FIGS. 10B-10F illustrate wick structures 140g, 140h, 140i, 140j, and 140k including different arrangements of the wick fibers 1010 around the central wick fiber 1020. In FIG. 10B, the wick structure 140g includes the wick fibers 1010 twisted together in the shape of a helix to form a bundle (e.g., a structure similar to a braided rope). In FIG. 10C, the wick structure 140h includes the wick fibers 1010 individually twisted and then twisted together in the shape of a helix to form a bundle. In FIG. 10D, the wick structure 140i includes the wick fibers 1010 longitudinally extending and arranged side by side.

[0101] In FIGS. 10E and 10F, the wick structures 140j and 140k are formed by twisting the wick fibers 1010 in the shape of a helix to form a bundle. Due to the difference in the angle of twist of the wires in the wick structures 140j and 140k, the two wick structures 140j and 140k have a different tensile force.

[0102] In some other embodiments, the central wick fiber 1020 is absent in the wick structures 140g, 140h, 140i, 140j, and 140k.

[0103] FIG. 10G is a cross-sectional view of a bundled wick structure including a plurality of wick fibers 1030 arranged in a circular manner. In an embodiment, the wick fibers 1030 include copper. However, the wick fibers 1030 can include other materials that facilitate the flow of condensed fluid by capillary force through the wick structure formed using the wick fibers 1030. FIG. 10H illustrates a wick structure 140m in which the plurality of wick fibers 1030 are loosely twisted together. FIG. 10I illustrates a wick structure 140n in which the plurality of wick fibers 1030 are longitudinally arranged. FIG. 10K illustrates a wick structure 140p in which the plurality of wick fibers 1030 are twisted together or in pairs to form a braided structure. In some other embodiments, the wick fibers 1010, 1020, and/or 1030 are flattened.

[0104] FIG. 11 illustrates a front perspective view of a vapor chamber 100i, according to embodiments of the disclosure. FIG. 12 illustrates an exploded view of the vapor chamber 100i, according to embodiments of the disclosure. FIG. 13 illustrates a plan view of a second casing 120i of the vapor chamber 100i in FIG. 11 including two wick structures, according to embodiments of the disclosure. FIG. 14 illustrates a plan view of a first casing 110i of the vapor chamber 100i in FIG. 11 including a sheet-like wick structure 130i, according to embodiments of the disclosure. The vapor chamber 100i also includes the working appendage 1150 (FIG. 2A) cooperatively formed by the first casing 110i and the second casing 120i.

[0105] Referring to FIGS. 11 and 12, the vapor chamber 100i includes a first casing 110i contacting a second casing 120i. As illustrated in FIGS. 11-14, the first casing 110i includes a first inner surface 183 and a first outer surface 185, and a mounting area 184 defined on the first outer surface 185. The first inner surface 183 is opposite to the first outer surface 185. The first outer surface 185 includes and forms part of the outer surface of the vapor chamber 100i and is substantially planar except for the mounting area 184. In some embodiments, the mounting area 184 is a raised (step) portion on the first outer surface 185. A first heat source H1 is thermally attached (or mounted) to the mounting area 184 and thereby to the outer surface of the vapor chamber 100i using a thermal paste or other techniques. The heat source H1 is not limited to any particular type of heat source H1 and can include heat sources such as electronic circuits (CPUs). Similarly, the shape and size of the mounting area 184 are not limited to any particular shape and size, and the mounting area 184 can have any desired shape and size on which the first heat source H1 can be correctly attached and from which the generated heat can be effectively dissipated. In addition, the location of the mounting area 184 is not limited to any particular location on the first outer surface 185, and the mounting area 184 can be located anywhere on the first outer surface 185 as long as the first heat source H1 can be correctly attached to the mounting area 184 and the first heat source H1 can effectively dissipate heat.

[0106] As illustrated in FIGS. 12 and 14, the sheet-like wick structure 130i is a generally planar, wing-shape structure extending along a length direction (X-direction in FIG. 11) of the vapor chamber 100i. The sheet-like wick structure 130i may be similar in some respects to the sheet-like wick structure 130 in FIG. 1, and therefore may be understood with reference thereto where like components are not described again in detail. The sheet-like wick structure 130i includes a first (or central) portion 85 including a first part B1a and a second part B1b adjacent and directly connected to the first part B1a in the length direction. The sheet-like wick structure 130i further includes a second portion B2 and a third portion B3, which form the opposite ends of the sheet-like wick structure 130i in the length direction. The first part B1a and the second part B1b are located between the second portion B2 and the third portion B3. As illustrated, the first part B1a is directly connected to the second portion B2 in the length direction and the second part B1b is directly connected to the third portion B3 in the length direction. Widths W2 of the second portion B2 and of the third portion B3 are about the same (or within a certain range of each other (for example, +/-1-5 mm)). Widths W1 of the first part B1a of the second part B1b are about the same (or

within a certain range of each other (for example, $\pm 1-5$ mm)). In some embodiments, the widths W2 of the second portion B2 and the third portion B3 are less than the widths W1 of the first part B1a and the second part B1b. In some other embodiments, the second portion B2 and the third portion B3 have different widths, and the first part B1a and second part B1b have different widths. However, even when the widths are different, the widths of the second portion B2 and the third portion B3 are smaller than the widths of the first part B1a and second part B1b.

[0107] The thermal resistances of the second portion B2 and the third portion B3 are about the same (or within a certain value of each other), and the thermal resistances of the first part B1a and second part B1b are about the same (or within a certain value of each other). In some embodiments, the thermal resistances of the second portion B2 and the third portion B3 are greater than the thermal resistances of the first part B1a and second part B1b. In other embodiments, the second portion B2 and the third portion B3 have different thermal resistances, and the first part B1a and second part B1b have different thermal resistances. However, even when the thermal resistances are different, the thermal resistances of the second portion B2 and the third portion B3 are greater than the thermal resistances of the first part B1a and second part B1b. Thus, the heat transfer performance (or ability) of the first part B1a and second part B1b is higher than the heat transfer performance of the second portion B2 and the third portion B3. In some embodiments, the thermal resistances of the second portion B2 and the third portion B3 are about the same (or within a certain value of each other), the thermal resistance of the first part B1a is less than the thermal resistances of the second portion B2 and the third portion B3, and the thermal resistance of the second part B1b is less than the thermal resistance of the first part B1a. Additionally, in other embodiments, the thermal resistance of the second part B1b is greater than the thermal resistance of the mounting area 184. Since the second part B1b is in thermal contact with the first heat source H1 attached to the mounting area 184, dissipation of heat generated from heat sources such as graphic processing units (GPUs) or central processing units (CPUs) is improved. It should be noted that, for the purposes of discussion herein, two parameters or quantities are considered to have a same value or a similar value when they have the same value or when a difference between their individual values is within an acceptable range, for example, $\pm 1-5$ units of measurement.

[0108] As illustrated in FIG. 13, the second casing 120i includes a second inner surface 283 and a second outer surface 285, and a second mounting area 284. The second inner surface 283 is opposite to the second outer surface 285. The second outer surface 285 includes and forms at least some of the outer surface of the vapor chamber 100i and is substantially planar. A second heat source H2 is attached in the second mounting area 284 to the outer surface of the vapor chamber 100i using a thermal paste, or other techniques. The second heat source H2 is illustrated as square shaped for the sake of illustration. It will be understood that the second heat source H2 is not limited to any particular shape and the second heat source H2 can have any desired shape.

[0109] The second casing 120i further includes an inner wick structure 140i, illustrated as being arranged in the second casing 120i. The inner wick structure 140i is a

generally planar, wing-shape structure extending along a length direction (X-direction in FIG. 13) of the vapor chamber 100i. The inner wick structure 140i may be similar in some respects to the sheet-like wick structure 130i in FIG. 12, and therefore may be understood with reference thereto where like components are not described again in detail. The wick structure 140i includes a first (or central) portion 95 including a first part A1a and a second part A1b adjacent to and directly connected to the first part A1a in the length direction. The wick structure 140i further includes a second portion A2 and a third portion A3, which form the opposite ends of the wick structure 140i in the length direction. The first part A1a and second part A1b are disposed between the second portion A2 and third portion A3. As illustrated, the first part A1a is directly connected to the second portion A2 in the length direction and the second part A1b is directly connected to the third portion A3 in the length direction. The first part A1a and second part A1b have a width W1, and the second portion A2 and the third portion A3 each have a width W2. Widths W2 of the second portion A2 and the third portion A3 are about the same (or within a certain predetermined range/tolerance of each other (for example, $\pm 1-5$ mm)) and the widths W2 of the first part A1a and second part A1b are about the same (or within a certain predetermined range/tolerance of each other (for example, $\pm 1-5$ mm)). The widths W2 of the second portion A2 and the third portion A3 are less than the widths W1 of the first part A1a and second part A1b. In other embodiments, the second portion A2 and the third portion A3 may have different widths, and the first part A1a and second part A1b may have different widths. However, even when the widths are different, the widths W2 of the second portion A2 and the third portion A3 are smaller than the widths W1 of the first part A1a and second part A1b.

[0110] A second heat source H2 is attached to the outer surface of the second casing 120i and thereby the outer surface of the vapor chamber 100i using a thermal paste, or other techniques. The second heat source H2 is illustrated as square shaped for the sake of illustration. It will be understood that the shape of the second heat source H2 is not limited to a square shape and the second heat source H2 can have any desired shape. In some embodiments, the thermal resistances of the second portion A2 and the third portion A3 are about the same (or within a certain value of each other, for example, ± 5 K/W of each other) and the thermal resistances of the first part A1a and second part A1b are about the same (or within a certain value of each other). In some embodiments, the thermal resistances of the second portion A2 and the third portion A3 are greater than the thermal resistances of the first part A1a and second part A1b. In other embodiments, the second portion A2 and the third portion A3 have different thermal resistances, and the first part A1a and second part A1b have different thermal resistances. However, even if the thermal resistances are different, the thermal resistances of the second portion A2 and the third portion A3 are greater than the thermal resistances of the first part A1a and second part A1b. Thus, the heat transfer performance (or ability) of the first part A1a and second part A1b is higher than the heat transfer performance (or ability) of the second portion A2 and the third portion A3. In some embodiments, the thermal resistances of the second portion A2 and the third portion A3 are about the same (or within a certain value of each other), the thermal resistance of the first part A1a is greater than the thermal resistances of

the second portion A2 and the third portion A3, and the thermal resistance of the second part A1b is greater than the thermal resistance of the first part A1a. Thus, the second part A1b that is in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i provides an improved heat transfer performance of heat generated from heat sources such as graphic processing units (GPUs) or central processing units (CPUs).

[0111] In some embodiments, the heat transfer performance of the second part B1b in thermal contact with the first heat source H1 attached to the outer surface of the first casing 110i is higher than the heat transfer performance of the second part A1b in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i. In some embodiments, the heat transfer performance of the second part B1b in thermal contact with the first heat source H1 attached to the outer surface of the mounting area 184 is about the same to the heat transfer performance of the second part A1b in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i.

[0112] In some embodiments, the sheet-like wick structure 130i of the first casing 110i and inner wick structure 140i of the second casing 120i are copper mesh wick structures, or other types of wick structures. In some embodiments, the thermal resistances of the first part A1a, second part A1b, second portion A2, and third portion A3 are similar to the thermal resistances of the first part B1a, second part B1b, second portion B2, and third portion B3, respectively. In some embodiments, the thermal resistances of the first part A1a, second part A1b, second portion A2, and third portion A3 are different than the thermal resistances of the first part B1a, second part B1b, second portion B2, and third portion B3, respectively. The thermal resistances can have any desired value as required by application or design provided the heat transfer performance of the second part B1b (that is in thermal contact with the first heat source H1) is higher than or equal to the heat transfer performance of the second part A1b (that is in thermal contact with the second heat source H2).

[0113] As illustrated in FIG. 12, the vapor chamber 100i includes a plurality of supporting structures 122 for supporting the first casing 110i and the second casing 120i and reducing buckling of the casings 110i and 120i. As a result, deformation of the vapor chamber 100i is reduced. In some embodiments, the supporting structures 122 are shaped as columns or pillars having a diameter greater than 0.4 mm but less than 50 mm, and height (or thickness) greater than 0.1 mm but less than 30 mm. However, embodiments are not limited in this regard. Other shapes and sizes are also possible provided the plurality of supporting structures 122 provide the required support to limit buckling of the first casing 110i and second casing 120i. In some embodiments, a copper sintered powdered wick layer 124 surrounds some of the supporting structures 122. In some embodiments, the copper sintered powdered wick layer 124 may be include, for example, oxygen-free copper, silicon-containing copper alloy, aluminum-containing copper alloy, a combination thereof, and the like. In other embodiments, the copper sintered powdered wick layer 124 includes porous wick structures. In some embodiments, the copper sintered powdered wick layer 124 surrounds the supporting structures 122 located in the first part A1a and second part A1b and corresponding first part B1a and second part B1b. However,

in other embodiments, the copper sintered powdered wick layer 124 surrounds all supporting structures 122.

[0114] The supporting structures 122 surrounded by the copper sintered powdered wick layer 124 reduce the thermal resistance of the first part A1a and second part A1b having the second heat source H2 attached to the outer surface of the second casing 120i and corresponding first part B1a and second part B1b having the first heat source H1 attached to the outer surface of the first casing 110i. As a result, the heat transfer performance is improved. In some embodiments, only some of the supporting structures 122 located in the first part A1a and second part A1b and the first part B1a and second part B1b are surrounded with the copper sintered powdered wick layer 124. In some embodiments, all the supporting structures 122 located in the first part A1a and second part A1b and corresponding first part B1a and second part B1b include the copper sintered powdered wick layer 124. The supporting structures 122 having the copper sintered powdered wick layer 124 reduce the thermal resistance of the first part A1a and the second part A1b having the second heat source H2 attached to the second casing 120i and the first part B1a and the second part B1b having the first heat source H1 attached to the first casing 110i, and thereby the heat transfer performance is improved.

[0115] Referring to FIGS. 12 and 13, illustrated are wick structures 141k and 142k disposed between the sheet-like wick structure 130i and the inner wick structure 140i, and each contacting contact the sheet-like wick structure 130i and the inner wick structure 140i. The wick structures 141k and 142k each have a generally serpentine shape (e.g., including multiple curves or turns) and are located in the gaps between the supporting structures 122.

[0116] The wick structure 141k includes a first section 111 that is disposed in the first part A1a. The wick structure 141k includes a second section 112 disposed in the second portion A2 and is inclined relative to the first section 111. In some embodiments, the second section 112 is inclined at an angle greater than the ideal horizontal (0°) and less than the ideal vertical (90°). The first section 111 is connected to the second section 112 at the connection between the first part A1a and second portion A2. However, embodiments are not limited thereto, and the first section 111 can be inclined relative to the second section 112 at angles or degrees of curvatures greater than or less than 45°.

[0117] The wick structure 142k includes a first section 113 that is disposed in the first part A1a and the second part A1b. The wick structure 142k includes a second section 114 disposed in the third portion A3 and is inclined relative to the first section 113. In some embodiments, the second section 114 is inclined at an angle greater than the ideal horizontal (0°) and less than the ideal vertical (90°). The first section 113 is connected to the second section 114 at the connection between the second part A1b and third portion A3. However, embodiments are not limited thereto, and the second section 114 can be inclined relative to the second section 114 at angles or degrees of curvatures greater than or less than 45°. The first section 111 of the wick structure 141k and the first section 113 of the wick structure 142k do not contact each other.

[0118] In some embodiments, both wick structures 141k and 142k overlap the heat source H2 located on the outer surface of the casing 120i. In some embodiments, the wick structures 141k and 142k overlap the heat source H2 via different sides (edges) of the heat source H2. In other

embodiments, the wick structure **142k** overlaps both heat sources **H1** and **H2**, and overlaps two or more sides (edges) of the heat source **H1**.

[0119] Referring to FIG. 13, the first section **111** includes an end **E1** of the wick structure **141k**. The end **E1** overlaps the heat source **H2** attached to the first casing **110i**. The first section **113** includes an end **E2** of the wick structure **142k**. The end **E2** overlaps the heat source **H2** attached to the first casing **110i**. The wick structure **142k** overlaps the heat source **H1** attached to the outer surface of the second casing **120i**.

[0120] The second section **112** includes an end **S1** of the wick structure **141k**. The second section **114** includes an end **S2** of the wick structure **142k**. In an example, the first and second heat sources **H1**, **H2**, are attached to the outer surfaces of the mounting area **184** of the first casing **110k** and second casing **120k**, in first part **A1a** and the second part **A1b**, respectively. The thermal resistance of the second part **A1b** is lower than the thermal resistance of the first part **A1a**. The thermal resistance of the first part **A1a** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** attached to the outer surface of the mounting area **184** is higher than the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2** attached to the outer surface of the second casing **120k**.

[0121] As illustrated, the end **S1** of the wick structure **141k** is located proximate (non-contacting) an edge **A32** of the second portion **A2**, and the end **S2** of the wick structure **142k** is located proximate (non-contacting) an edge **A33** of the third portion **A3** opposite the edge **A32**. The second sections **112** and **114** are disposed proximate (non-contacting) an edge **A31** of the second casing **120i** that forms (or otherwise defines) the first part **A1a**, the second part **A1b**, the second portion **A2**, and the third portion **A3** of the second casing **120k**. Referring to FIG. 13, the edge **A31** forms the lower edge of the second casing **120i** and connects the edge **A32** and edge **A33**. In other embodiments, the end **S1** may contact the edge **A32** and the end **S2** may contact the edge **A33**. In yet other embodiments, the second sections **112** and **114** contact the edge **A32**.

[0122] As illustrated, the end **E1** of the wick structure **141k** and the end **E2** of the wick structure **142k** are separated (non-contacting) from the edge **A31**. The end **E2** of the wick structure **142k** is located closer to the edge **A31** than the end **E1** of the wick structure **141k** and the first sections **111** and **113** (and thereby the ends **E1** and **E2**) are separated from each other either vertically (as illustrated) or horizontally. In other embodiments, the end **E2** of the wick structure **142k** is closer in proximity to the edge **A31** than the end **E1** of the wick structure **141k**. In some embodiments, the ends **E1** and **E2** cross each other in the vertical plane, for example, the YZ plane in FIG. 13 and the ends **E1** and **E2** are vertically separated from each other and overlapping in the horizontal plane, for example, the XY plane in FIG. 13. However, in other embodiments, the ends **E1** and **E2** contact each other in the vertical plane, for example, the YZ plane in FIG. 13, and, in the horizontal plane, for example, the XY plane, the ends **E1** and **E2** are either offset from each other or are aligned with each other. The first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110i** is overlapped by the wick structure **142k**, and both the wick structure **141k** and the wick structure **142k** overlaps the

second heat source **H2** attached to the outer surface of the second casing **120k**. In some embodiments, the heat source **H2** has a relatively higher power consumption than heat source **H1**. In some embodiments, the ends **E1** and **E2** of the wick structures **141k** and **142k** are located within the boundaries (edges) of the second heat source **H2**.

[0123] It will be understood that the location of the mounting area **184** may be varied depending on the application and design requirements, as long as the first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110i** is overlapped by the wick structure **142k**.

[0124] The wick structures **141k** and **142k** decrease a distance the cooling fluid has to circulate in the vapor chamber. The length of the wick structures **141k** and **142k** are not limited to any particular length and may be adjusted according to user and design requirements.

[0125] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the first heat source **H1** and/or second heat source **H2** are (is) located at a level higher than the second portion **A2** and third portion **A3**), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the first and/or second heat sources **H1** and/or **H2**, respectively. The thermal resistance of the second part **A1b** is lower than the thermal resistance of the first part **A1a**. The thermal resistance of the first part **A1a** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** attached to the outer surface of the mounting area **184** is higher than the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2** attached to the outer surface of the second casing **120i**.

[0126] In some embodiments, the thermal resistance of the second part **B1b** is higher than the thermal resistance of the mounting area **184**. In some embodiments, the supporting structures **122** having the copper sintered powdered wick layer **124** surrounding the supporting structures **122** within the first part **A1a** and second part **A1b**, further reduces the thermal resistance of the first part **A1a** and second part **A1b** having the first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110k** and second heat source **H2** attached to the outer surface of the second casing **120i**, further improving the heat transfer performance. The cooling fluid (in vapor state) flows toward the second portion **A2** and the third portion **A3** and turns back to liquid state, and the cooling fluid can flow to the first part **A1a** and second part **A1b** via the wick structures **141k** and **142k**. This configuration of the wick structures **141k** and **142k**, the portions of different thermal portions, and the copper sintered powered wick layers surrounding the supporting structures, decreases a temperature difference between the first part **A1a** and second part **A1b** and the second portion **A2** and the third portion **A3** by around 15 degrees Celsius or higher. The cooling fluid continuously circulates between the first part **A1a** and second part **A1b** and second portion **A2** and third portion **A3**, thereby dissipating the heat generated by the first and/or second heat sources **H1** and **H2**.

[0127] Compared to the embodiment of FIG. 8A, the wick structures **141k** and **142k** each have a generally flowing, sloping-ramped-shape and do not contact each other. Portions of the sheet-like wick structure **130i** contacting the first

casing 110i may have different thermal resistances. An inner wick structure 140i of the second casing 120i is included, also having different and corresponding thermal resistances to the sheet-like wick structure 130i. In some embodiments, the thermal resistances of the first part A1a, second part A1b, second portion A2, and third portion A3 of the second casing 120k are similar to the thermal resistances of the first part B1a, second part B1b, second portion B2, and third portion B3 of the sheet-like wick structure 130k, respectively. The thermal resistances of the first part A1a, second part A1b, second portion A2, and third portion A3 and first part B1a, second part B1b, second portion B2, and third portion B3 can be varied as required by application and design. A mounting area 184 includes a thickness and the mounting area 184 has a lower thermal resistance than that of the second part B1b, first part B1a, second portion B2, and third portion B3, respectively. A high temperature heat source, such as a processor (CPU, GPU, and the like) may be attached to the outer surface of the mounting area 184 of the first casing 110i. The thermal resistance of the second part A1b is lower than the first part A1a, and the thermal resistance of the first part A1a is lower than the thermal resistances of the second portion A2 and third portion A3, respectively. An additional heat source, such as a CPU may be attached to the outer surface of the second casing 120i opposite the first part A1a.

[0128] The wick structures 141k and 142k, portions of different thermal resistances, and the copper sintered powered wick layers surrounding the supporting structures 122, further improves circulation of cooling fluid in the vapor chamber 100i, for example, against gravity. Thus, the efficiency of the vapor chamber in dissipating heat is further improved.

[0129] In the embodiments above, the first part A1a and second part A1b of the inner wick structure 140i and corresponding first part B1a and second part B1b of the sheet-like wick structure 130i are disclosed as separate components positioned adjacent side by side each other. However, in other embodiments, a wick structure is arranged within another wick structure.

[0130] FIG. 15 illustrates a plan view of a second casing 120i including two wick structures, according to embodiments of the disclosure. FIG. 16 illustrates a plan view of a first casing including a sheet-like wick structure according to embodiments of the disclosure. The second casing 120i having an inner wick structure 140i may be used in the vapor chamber 100i that includes a first casing 110i and a sheet-like wick structure 130i.

[0131] As illustrated in FIG. 15, the second casing 120i having the inner wick structure 140i, is generally wing-shape including a first part A1c, a second part A1d positioned within the first part A1c, a second portion A2, and a third portion A3, wherein the first part A1c and second part A1d are positioned between the second portion A2 and third portion A3. The first part A1c has a width W1, the second part A1d has a width W3, and the second portion A2 and the third portion A3 each have a width W2. In some embodiments, the widths W2 of the second portion A2 and the third portion A3 are about the same, and the widths W2, W3 of the first part A1c and second part A1d are different. In some embodiments, the widths W2 of the second portion A2 and the third portion A3 are smaller than the widths W1, W3 of the first part A1c and second part A1d. In other embodiments, the second portion A2 and the third portion A3 have

different widths and the first part A1c and second part A1d may have different widths, but the widths W2 of the second portion A2 and the third portion A3 are smaller than the widths W1, W3 of the first part A1c and second part A1d. As illustrated, the first part A1c completely surrounds the second part A1d. In other words, the second part A1d is located entirely within the first part A1c.

[0132] As illustrated in FIG. 16, the sheet-like wick structure 130i is generally wing-shaped including a first part B1c, a second part B1d within the first part B1c, a second portion B2, and a third portion B3, wherein the first part B1c and second part B1d are positioned between the second portion B2 and third portion B3. As illustrated, the first part B1c completely surrounds the second part B1d. In other words, the second part B1d is located entirely within the first part B1c. In some embodiments, the widths of the second portion B2 and the third portion B3 are about the same and the widths of the first part B1c and second part B1d are different. In some embodiments, the widths of the second portion B2 and the third portion B3 are smaller than the width of the second part B1d, and the width of the second part B1 is smaller than the width of the first part B1c. In other embodiments, the second portion B2 and the third portion B3 have different widths, and the first part B1c and second part B1d have different widths, but the widths of the second portion B2 and the third portion B3 are smaller than the widths of the first part B1c and second part B1d.

[0133] In some embodiments, the thermal resistances of the second portion B2 and the third portion B3 are about the same, and the thermal resistances of the first part B1c and second part B1d are different. In some embodiments, the thermal resistances of the second portion B2 and the third portion B3 are higher than the thermal resistances of the first part B1c and second part B1d. In other embodiments, the second portion B2 and the third portion B3 have different thermal resistances, and the first part B1c and second part B1d have different thermal resistances, but the thermal resistances of the second portion B2 and the third portion B3 are greater than the thermal resistances of the first part B1c and second part B1d. Thus, the heat transfer performance of the first part B1c and second part B1d is higher than the heat transfer performance of the second portion B2 and the third portion B3. In other embodiments, the thermal resistances of the second portion B2 and the third portion B3 are about the same, the thermal resistance of the first part B1c is less than the thermal resistances of the second portion B2 and the third portion B3 and the thermal resistance of the second part B1d is less than the thermal resistance of the first part B1c. In still other embodiments, the thermal resistance of the second part B1d is higher than the thermal resistance of the mounting area 184. Thus, the second part B1d in thermal contact with the first heat source H1 attached to the outer surface of the mounting area 184 dissipates heat more efficiently, and can be used to dissipate heat generated by high temperature heat sources, such as, central processing units (CPUs), graphic processing units (GPUs), and the like.

[0134] Referring to FIG. 15, the second heat source H2 is attached to the outer surface of the second casing 120i and thereby the outer surface of the vapor chamber 100i using a thermal paste, or other techniques. In some embodiments, the thermal resistances of the second portion A2 and the third portion A3 are about the same, and the thermal resistances of the first part A1c and second part A1d are different. In some embodiments, the thermal resistances of

the second portion A2 and the third portion A3 are higher than the thermal resistances of the first part A1c and second part A1d. In other embodiments, the second portion A2 and the third portion A3 have different thermal resistances, and the first part A1c and second part A1d have different thermal resistances, but the thermal resistances of the second portion A2 and the third portion A3 are higher than the thermal resistances of the first part A1c and second part A1d. Thus, the heat transfer performance of the first part A1c and second part A1d is higher than the heat transfer performance of the second portion A2 and the third portion A3. In other embodiments, the thermal resistances of the second portion A2 and the third portion A3 are about the same, the thermal resistance of the first part A1c is lesser than the thermal resistances of the second portion A2 and the third portion A3, and the thermal resistance of the second part A1d is lesser than the thermal resistance of the first part A1c. Thus, the second part A1d in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i dissipates heat more efficiently, and can be used to dissipate heat generated by high temperature heat sources such as central processing units (CPUs), graphical processing units (GPUs), and the like.

[0135] In some embodiments, the heat transfer performance of the second part B1d in thermal contact with the first heat source H1 attached to the mounting area 184 is about the same as the heat transfer performance of the second part A1d in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i. In some embodiments, the heat transfer performance of the second part B1d in thermal contact with the first heat source H1 attached to the mounting area 184 is higher than the heat transfer performance of the second part A1d in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i.

[0136] In some embodiments, the sheet-like wick structure contacting the first casing and inner wick structure of the second casing are copper mesh wick structures, or other types of wick structures. In some embodiments the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 are similar to the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3, respectively. In some embodiments, the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 are about the same as the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3, respectively. However, in other embodiments, the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 are different from the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3, respectively. The thermal resistances can have any desired value as required by application or design provided, the heat transfer performance of the second part B1d in thermal contact with the first heat source H1 attached to the outer surface of the mounting area is relatively higher than or equal to the heat transfer performance of the second part in thermal contact with the second heat source H2 attached to the outer surface of the second casing.

[0137] The wick structures 141k and 142k, each, contact the sheet-like wick structure 130i and the inner wick structure 140i of the second casing 120i. The wick structures

141k and 142k decrease a distance the cooling fluid has to circulate in the vapor chamber during operation of the vapor chamber 100i.

[0138] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the first heat source H1 and/or second heat source H2 are (is) located at a level higher than the second portion A2 and third portion A3), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the first and/or second heat sources H1 and/or H2, respectively. The thermal resistance of the second part A1d is lower than the thermal resistance of the first part A1c. The thermal resistance of the first part A1c is lower than the thermal resistances of the second portion A2 and third portion A3. The heat transfer performance of the second part B1d in thermal contact with the first heat source H1 is similar to the heat transfer performance of the second part A1d in thermal contact with the second heat source H2.

[0139] In some embodiments, the thermal resistance of the second part B1d is greater than the thermal resistance of the mounting area 184. In some embodiments, the supporting structures 122 within the first part A1c and second part A1d are surrounded with the copper sintered powdered wick layer 124, and this reduces the thermal resistance of the first part A1c and second part A1d having the first heat source H1 and second heat source H2. As a result, heat transfer performance is increased. The cooling fluid (in vapor state) flows toward the second portion A2 and the third portion A3 and turns returns to liquid state, and the cooling fluid can flow to the second part A1d via the wick structures 141k and 142k. This configuration of the wick structures 141k and 142k, the portions of different thermal resistances, and the copper sintered powdered wick layers surrounding the supporting structures decreases a temperature difference between the second part A1d and the second portion A2 and the third portion A3 by about 15 degrees Celsius or higher. The cooling fluid continuously circulates between the second part A1d, first part A1c and second portion A2 and third portion A3, thereby dissipating the heat generated by the first and/or second heat sources H1 and/or H2, respectively.

[0140] The wick structures 141k and 142k each have a flowing, generally sloping-ramped-shape and do not contact each other. Portions of the sheet-like wick structure 130i contacting the first casing 110i may have different thermal resistances. An inner wick structure 140i of the second casing 120i also has different thermal resistance compared to the sheet-like wick structure 130i. In some embodiments, the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 of the second casing 120k are similar to the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3 of the sheet-like wick structure 130i, respectively. The thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 and first part B1c, second part B1d, second portion B2, and third portion B3 may be changed as required by application and design for obtaining a desired performance. A second heat source H2 may be attached to the outer surface of the second casing 120i in addition to a first heat source H1 being attached to the outer surface of the first casing 110i. A mounting area 184 includes a thickness and the mounting area 184 has a lower thermal resistance than that of the second part B1d, first part B1c, second portion B2, and third portion B3, respectively. A high temperature heat source, such as a

central processing unit (CPU), graphical processing unit (GPU), and the like may be attached to the outer surface of the mounting area **184** of the first casing **110i**. The thermal resistance of the second part **A1d** is lower than the first part **A1c**, and the thermal resistance of the first part **A1c** is lower than the thermal resistances of the second portion **A2** and third portion **A3**, respectively. One or more high temperature heat sources, may be attached to the outer surface of the second casing **120i** opposite the second part **A1d**. The wick structure **142k** overlaps the second heat source **H2** and first heat source **H1**. The wick structures **141k** and **142k**, varied thermal resistance portions, and the copper sintered powered wick layers surrounding the supporting structures **122** improve circulation of cooling fluid in the vapor chamber **100i**. Thus, the efficiency of the vapor chamber in dissipating heat is further increased.

[0141] Instead of the two wick structure some embodiments include a single wick structure. FIG. 17 illustrates a plan view of the second casing **120i** including a wick structure **141m**, according to embodiments of the disclosure.

[0142] The wick structure **141m** contacts the sheet-like wick structure **130i** contacting the first casing **110i** and inner wick structure **140i** of the second casing **120i**. The wick structure **141m** is a generally bell shape structure that is flattened at the top and is located in the gaps between the supporting structures **122**.

[0143] The wick structure **141m** includes a first section **131** that is disposed in the first part **A1a** and second part **A1b**. The wick structure **141m** includes a second section **112** disposed in the second portion **A2** and is located at a lower level (referring to the orientation in FIG. 17) than the first section **131**. The second section **112** is arranged relatively horizontal and portions of the first section **131** that are connected to the second section **112** are inclined (e.g., positioned at an angle greater than 0° and less than 90°) relative to the second section **112**. The first section **131** is connected to the second section **112** at the connection between the first part **A1a** and second portion **A2**. As illustrated, the wick structure **141m** generally follows the shape of the casing **120i**. In an example, the first section **131** meets the second section **112** at angles or degrees of curvatures greater than or less than 45°, and greater than 0° and 90°.

[0144] The wick structure **141m** includes a third section **104** disposed in the third portion **A3** and is located at a lower level (referring to the orientation in FIG. 17) than the first section **131**. The third section **104** is arranged relatively horizontal and portions of the first section **131** that are connected to the third section **104** are inclined (e.g., positioned at an angle greater than 0° and less than 90°) relative to the third section **104**. In some embodiments, the second section **112** and the third section **104** are located at a same horizontal level (e.g., collinear). In some other embodiments, the second section **112** and the third section **104** are vertically offset from each other. The first section **113** is connected to the third section **104** at the connection between the second part **A1b** and third portion **A3**. In an example, the first section **113** meets the third section **104** at angles or degrees of curvatures greater than or less than 45°, and greater than 0° and 90°. The first section **131** of the wick structure **141m** and the second section **104** of the wick structure **141m** form the opposite ends of the wick structure **141m** and do not contact each other.

[0145] The first section **131**, located in the first part **A1a** and second part **A1b**, overlaps the first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110i** and the second heat source **H2** attached to the outer surface of the second casing **120i**. The second section **112** includes an end **S3** of the wick structure **141m** in the second portion **A2**. The third section **104** includes an end **S4** of the wick structure **141m** in the third portion **A3**. In an example, the first and second heat sources **H1**, **H2**, are attached to the outer surfaces of the mounting area **184** of the first casing **110i** and second casing **120i**, in the second part **A1b** and first part **A1a**, respectively. The thermal resistance of the second part **A1b** is lower than the thermal resistance of the first part **A1a**. The thermal resistance of the first part **A1a** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** is higher than the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2**.

[0146] As illustrated, the end **S3** of the wick structure **141m** is located proximate (non-contacting) an edge **A32** of the second portion **A2** and the end **S4** of the wick structure **141m** is located proximate (non-contacting) an edge **A33** of the third portion **A3** opposite the edge **A32**. The second sections **112** and **114** are disposed proximate (non-contacting) an edge **A31** of the second casing **120i** that forms (or otherwise defines) the first part **A1a**, the second part **A1b**, the second portion **A2**, and the third portion **A3** of the second casing **120i**. The edge **A31** connects the edge **A32** and edge **A33**. In other embodiments, the end **S3** may contact the edge **A32** and the end **S4** may contact the edge **A33**.

[0147] As illustrated, the first section **131** of the wick structure **141m** is located closer to the top edge **A34** of the second casing **120i** (and thereby the top edge of the vapor chamber **100i**) than the edge **A31**. In other embodiments, the first section **131** of the wick structure **141m** is located closer to the edge **A31** than the edge **A34**. The placement of the first section **131** is not limited in any regard, and the first section **131** can be located anywhere in the first part **A1a** and second part **A1b** provided the first section **131** (and thereby the wick structure **141m**) overlaps the first heat source **H1** and second heat source **H2**.

[0148] It will be understood that the location of the mounting area **184** may be varied depending on the application and design requirements. The mounting area **184** can be located at any desired location provided the first heat source **H1** is overlapped by the wick structure **141m**.

[0149] The wick structure **141m** decreases a distance the cooling fluid has to circulate in the vapor chamber **100i**. The length of the wick structure **141m** is not limited to any particular length and may be adjusted according to application and design requirements.

[0150] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the first heat source **H1** and/or second heat source **H2** are located at a level higher than the second portion **A2** and third portion **A3**), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the first and/or second heat sources **H1** and/or **H2**, respectively. The thermal resistance of the second part **A1b** is lower than the thermal resistance of the first part **A1a**. The thermal resistance of the first part **A1a** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer

performance of the second part B1*b* in thermal contact with the first heat source H1 attached to the outer surface of the mounting area 184 is greater than the heat transfer performance of the second part A1*b* in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120*m*. In some embodiments, the thermal resistance of the second part B1*b* is higher than the thermal resistance of the mounting area 184. In some embodiments, the supporting structures 122 are surrounded by the copper sintered powdered wick layer 124 in the first part A1*a* and second part A1*b*, and this further decreases the thermal resistance of the first part A1*a* and second part A1*b*, and thereby increases the heat transfer performance. The cooling fluid (in vapor state) flows toward the second portion A2 and the third portion A3 and turns back to liquid state, and the cooling fluid can flow to the first part A1*a* and second part A1*b* via the wick structure 141*m*. This configuration of the wick structure 141*m*, the portions having different thermal resistances, and the copper sintered powdered wick layers surrounding the supporting structures, decreases a temperature difference between the first part A1*a* and second part A1*b* and the second portion A2 and the third portion A3 by around 15 degrees Celsius or more. The cooling fluid continuously circulates between the first part A1*a* and second part A1*b* and second portion A2 and third portion A3, thereby dissipating the heat generated by the first and/or second heat sources H1 and/or H2, respectively.

[0151] FIG. 18 illustrates a plan view of the second casing 120*i* including the wick structure 141*m*, according to embodiments of the disclosure. In this embodiment, the second casing 120*i* includes the inner wick structure 140*i* including the first part A1*c*, the second part A1*d* positioned within the first part A1*c*, the second portion A2, and the third portion A3. The first part A1*c* and second part A1*d* are positioned between the second portion A2 and third portion A3. The structure of the second casing 120*i* including the wick structure 141*m* is similar in some respects to the second casing 120*i* illustrated in FIG. 15 including wick structures 141*k* and 142*k*, and will be understood with reference thereto where like reference numerals indicate like components and are not described again in detail.

[0152] The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A heat dissipation device, comprising:

a first casing including:

a first inner surface having a first wick structure, the first wick structure including:

a first portion;
a second portion; and
a third portion,

wherein the first portion is disposed between the second portion and third portion;

a first outer surface opposite the first inner surface; and
a first mounting plate for receiving a first heat source disposed on the first outer surface and located overlapping the first portion of the first wick structure;
a second casing coupled to the first casing, the second casing including:
a second inner surface having a second wick structure, the second wick structure including:
a first portion opposite the first portion of the first wick structure;
a second portion opposite the second portion of the first wick structure; and
a third portion opposite the third portion of the first wick structure,
wherein the first portion of the second wick structure is disposed between the second portion and third portion of the second wick structure; and
a second outer surface opposite the second inner surface and including a second mounting plate different from the first mounting plate for receiving a second heat source, the second mounting plate located overlapping the first portion of the second wick structure;
a third wick structure having at least two bends and arranged between the first inner surface and the second inner surface and arranged in the first portions of the first wick structure and the second wick structure, the second portions of the first wick structure and the second wick structure, and the third portions of the first wick structure and the second wick structure, the third wick structure at least partially overlaps the second mounting plate and the first mounting plate; and
a plurality of columns arranged between the first and second inner surfaces,
wherein thermal resistances of the first portion of the first wick structure, second portion of the first wick structure and third portion of the first wick structure are same as thermal resistances of the first portion of the second wick structure, second portion, and third portion, respectively,
the thermal resistance of the first portion of the first wick structure is less than the thermal resistances of the second portion and the third portion of the first wick structure, and
the thermal resistance of the first portion of the second wick structure is less than the thermal resistances second portion and third portion of the second wick structure.

2. The heat dissipation device of claim 1, wherein a thermal resistance of the first mounting plate is less than the thermal resistances of the first portion of the first wick structure and the first portion of the second wick structure.

3. The heat dissipation device of claim 1, wherein the first portion of the second wick structure comprises a first part and a second part and the first portion of the first wick structure comprises a first part and a second part, the first part of the first wick structure is opposite the first part of the second wick structure, and the second part of the first wick structure is opposite the second part of the second wick structure.

4. The heat dissipation device of claim 3, wherein the first part and second part, each of the first wick structure are adjacent each other,

the first part and second part, each of the second wick structure are adjacent each other,

the second portion of the first wick structure is adjacent the first part of the first wick structure,
 the second portion of the second wick structure is adjacent the first part of the second wick structure,
 the third portion of the first wick structure is adjacent the second part of the first wick structure,
 the third portion of the second wick structure is adjacent the second part of the second wick structure, and
 a thermal resistance of the second part of the first wick structure is less than the thermal resistance of the first part of the first wick structure and a thermal resistance of second part of the second wick structure is less than the thermal resistance of the first part of the second wick structure and
 the second mounting plate is disposed overlapping the first part of the first wick structure and the first part of the second wick structure and the first mounting plate is disposed overlapping the second part of the first wick structure and second part of the second wick structure.

5. The heat dissipation device of claim 3, wherein the first part of the first wick structure is within the second part of the first wick structure,
 the first part of the second wick structure is within the second part of the second wick structure,
 the second portion of the first wick structure is adjacent the second part of the first wick structure,
 the second portion of the first wick structure is adjacent to the second part of the second wick structure,
 the third portion of the first wick structure is adjacent the second part of the first wick structure,
 the third portion of the second wick structure is adjacent the second part of the second wick structure, and

wherein a thermal resistance of the first part of the second wick structure are less than a thermal resistance of the first part of the first wick structure,

a thermal resistance of the second part of the first wick structure is less than a thermal resistance of the second part of the first wick structure, and

the second mounting area and first mounting area are disposed overlapping the first part of the second wick structure and the first part of the first wick structure, respectively.

6. The heat dissipation device of claim 5, wherein a cross-sectional area of the first part of the second wick structure is greater than a cross-sectional area of the second part of the second wick structure, and

a cross-sectional area of the second part of the second wick structure is greater than cross-sectional areas of the second portion and third portion of the second wick structure.

7. The heat dissipation device of claim 1, wherein each of the plurality of columns adjacent to the third wick structure are surrounded by porous wick structures.

8. The heat dissipation device of claim 1, wherein a cross-sectional area of the first portion of the first wick structure is greater than cross-sectional areas of the second portion and the third portion of the first wick structure.

9. The heat dissipation device of claim 1, wherein each of the plurality of columns overlapping the second mounting plate and first mounting plate are surrounded by porous wick structures.

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